Supplemental Material for "Emergent $\mathcal{PT}$ Symmetry and Exceptional Points in a Driven Dirac Impurity"

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This Supplemental Material provides the technical derivations and validation checks that support the self-contained main article. It includes the auxiliary-mode influence functional, Keldysh mean-field equations, physical fluctuation–dissipation relation (FDR) construction, number continuity, shifted-resonance condition, the complete biorthogonal and finite-U charge resolvents, one-particle scattering-phase checks, the controlled branch-wise linearization, the exact finite-U two-electron contact algebra, the full-momentum two-cut nonlinear-pole exceptional-point (EP) certificate, its finite-U matrix block-Wilson continuation, and numerical validation figures. Throughout, $\mathcal{PT}$ denotes parity–time symmetry.

NOTATION AND PARAMETER CONVENTIONS

$t_{\rm av}, \tau, T_{\rm b}$**:** Wigner center time $(t+t')/2$, relative time $t-t'$, and reservoir temperature.

η, ζ_σ**:** Chiral bath index $\eta = \pm$ and spin sign $\zeta_\uparrow = +1$, $\zeta_\downarrow = -1$.

λ, j_α**:** Spin–orbit coupling (SOC) strength and eigenvalues of the dimensionless exchange matrix $\rho_b J^{\rm bio}$; $j_{\rm scr}$ is the largest positive real part.

$\alpha_{\rm b}, k_{\rm max}, D_{\rm uv}$**:** Quadratic bath-dispersion coefficient, radial momentum cutoff, and ultraviolet energy cutoff. The normalized $k = 0$ core figures contain neither cutoff; $k_{\rm max} = \pi/4$ is used in the causal gate and $D_{\rm uv} = \pi/4$ only in the sharp-cutoff reachability audit.

$W_k, W, \Lambda_\eta^{\rm PV}$**:** Static bare impurity–bath hybridization and principal-value bath shift. The drive does not set W_k; at the saddle the vertex is rW_k.

$\beta_0, \tilde{\beta}$**:** Signed frozen drive-control amplitude and nonnegative slave-boson mean-field (SBMF) projection $\tilde{\beta} = |\beta_0||b_c|$. On the audited path $D = 0.10$ and $G = 0.20\beta_0$, so β_0 is not itself a decay rate. The local core and the SOC-degenerate $k = 0$ bath-dressed condition coincide exactly at $\beta_{0,\rm core}^{\rm EP} = 0.50$. The nonzero-$k$ root search is boundary-limited at $\beta_0 = 0.55$ and establishes no interior projected EP.

s**:** Impurity pole splitting; ζ_σ is used for the spin sign.

s_a, δ_a**:** Eigenvalues $\mathsf{s}_a = e^{2i\delta_a}$ and eigenphases of the complete unitary scattering matrix. They are distinct from the EP pole splitting s and from the fixed chiral labels $\eta = \pm$.

$D, G, \Gamma_{\rm PT}, \bar{\Gamma}$**:** Constant-core coherent splitting, constant-core relative channel, projected relative decay imbalance, and common bath width. The symbols $\tilde{\Gamma}_\eta$ and $\Gamma_\sigma^{\rm net}$ denote, respectively, the chiral frozen-kernel width and the physical net channel width.

ω, Ω, ω_0**:** Fourier frequency, drive frequency, and a selected spectral pole or resonance frequency; none of these denotes temperature.

$\mathcal{E}_n, E_{\rm EP}, \Lambda(\omega)$**:** Non-Hermitian eigenvalues, EP eigenvalue, and real bath self-energy.

$\mathcal{P}, \mathfrak{a}, \mathfrak{b}$**:** Full-kernel Schur polynomial and the two local contact denominators; these are distinct from the bath coefficient $\alpha_{\rm b}$.

$\mathcal{F}_{\rm dist}, \mathcal{F}_{\rm MF}$**:** Distribution matrix and mean-field free-energy functional.

For clarity, the same constant two-component matrix is written in three conventions:

representation	constant matrix	identification
spin basis	$D\sigma_x + iG\sigma_z$	$D = \Delta_{\rm eff}$, $G = \Gamma_{\rm PT}$
fixed chiral basis	$D\sigma_z + iG\sigma_x$	Hadamard rotation
companion algebra	$\gamma\sigma_x + i\beta\sigma_z$	$\gamma = D$, $\beta = G$

The frozen drive coordinate is instead β_0, with $G = 0.20\beta_0$ on the audited path; it must not be identified with the matrix coefficient G or with the companion symbol β. The normalized geometry plots use $D = 0.50$ and $G = |\beta_0|$, whereas the small-width causal/renormalization-group (RG) audit uses $D = 0.10$ and $G = 0.20\beta_0$. Both place the dimensionless core crossing at $|\beta_0| = 0.50$ because all constant-core coefficients are rescaled together; their displayed β_0 axes are a shared control-coordinate normalization, not identical physical decay rates.

Appendix A: Impurity-block EP condition and condition-number scaling

This appendix records the impurity-block algebra used in the main text. The Hadamard rotation itself is shown in Eqs. (52)–(53); here we keep only the EP and condition-number scaling that underlies the diagnostic interpretation.

The frozen impurity block is $h_d = \left[\begin{smallmatrix} \tilde{\epsilon}_\xi + D & iG \\ iG & \tilde{\epsilon}_\xi - D \end{smallmatrix}\right]$, with eigenvalues $E_\pm = \tilde{\epsilon}_\xi \pm s$ and splitting $s = \sqrt{D^2 - G^2}$. The regimes are $\mathcal{PT}$-unbroken ($D > G$), exceptional ($D = G$), and $\mathcal{PT}$-broken ($G > D$) for positive parameters. Right eigenvectors may be chosen as $|r_\pm\rangle \propto [G,\, i(D \mp s)]^T$, so

$$\det R = 2iGs, \qquad R = (|r_+\rangle, |r_-\rangle). \tag{A1}$$

Thus, near the bare EP,

$$\kappa_{\rm imp} = \|R\|\|R^{-1}\| \sim \frac{C}{|s|}, \tag{A2}$$

with a nonsingular constant C. The standard local U does not cut off this algebraic divergence on the equal-velocity, scalar-hybridization linearized submanifold because it preserves the complexified pseudospin symmetry. Physical observables built from the complete resolvent remain finite. Curvature, unequal velocities, frequency dependence, or component-dependent vertices can lift the exact symmetry and must be analyzed separately.

1. Slave-boson scaling and the core EP

The bare impurity–bath hybridization W in Eq. (3) of the main article is time independent. At the saddle it becomes rW. Since the projected impurity self-energy is $\Sigma_d = W^\dagger G_c W$, every induced impurity channel carries the same factor r^2:

$$\Gamma_{\rm PT} = r^2 \Gamma^{(0)}_{\rm PT}(\beta_0), \quad \Delta_{\rm eff} = r^2 \Delta^{(0)}_0, \quad \bar{\Gamma} = r^2 \Gamma_0. \tag{A3}$$

Consequently

$$h^2_{\rm rel} = r^4 \left\{ [\Delta^{(0)}_0]^2 - [\Gamma^{(0)}_{\rm PT}(\beta_0)]^2 \right\} I. \tag{A4}$$

The common r^4 cancels from the local core condition for every $r > 0$. At $r = 0$ the result is an ordinary decoupled degeneracy. A bath-dressed finite-k minimum can drift because the SOC-resolved Schur terms do not remain a common scalar shift; the audited search reaches the scan boundary and is therefore not classified as an EP. At $k = 0$, by contrast, $\varepsilon_{c,+}(0) = \varepsilon_{c,-}(0) = 0$, and the Schur polynomial factorizes exactly as

$$\mathcal{P}(z, 0) = \left(z - \tilde{\epsilon}_\xi - \frac{r^2 W^2}{z} \right)^2 - r^4 \left\{ [\Delta^{(0)}_0]^2 - [\Gamma^{(0)}_{\rm PT}(\beta_0)]^2 \right\}. \tag{A5}$$

The impurity-related $k = 0$ double-root condition is therefore exactly the local core condition for arbitrary W and $r > 0$.

Appendix B: Angular Harmonic Decomposition and 2D→1D Reduction

We show how the 2D Dirac bath with trigonal anisotropy reduces to an effective 1D problem after integrating out all angular-momentum harmonics $m \neq 0$. This Gaussian reduction is the microscopic origin of the 1D BA structure in Sec. IX.

1. Angular Fourier Decomposition

In 2D polar momentum coordinates (k, θ), the bath fermion is decomposed into angular-momentum channels:

$$c_{k,\theta,\sigma} = \frac{1}{\sqrt{2\pi}} \sum_{m=-\infty}^{\infty} c_{k,m,\sigma}\, e^{im\theta}. \tag{B1}$$

The impurity at position $\mathbf{r} = 0$ (the lattice site) is local, so only the isotropic ($m = 0$) channel hybridizes with it:

$$c_{\mathbf{r}=0,\sigma} = \sum_k \frac{1}{\sqrt{2\pi}}\, c_{k,0,\sigma}. \tag{B2}$$

Higher harmonics $m \neq 0$ are decoupled from the impurity in the absence of anisotropy.

2. C_3 Selection Rule from $\cos 3\theta$

The trigonal anisotropy $\beta_3(k) k^3 \cos 3\theta$ is expanded as $\cos 3\theta = \frac{1}{2}(e^{3i\theta} + e^{-3i\theta})$. Substituting Eq. (B1) and performing the θ integral: $\int_0^{2\pi} \frac{d\theta}{2\pi}\, e^{i(m'-m)\theta}\, e^{\pm 3i\theta} = \delta_{m',m\mp 3}$. The trigonal term therefore couples only channels differing by $\Delta m = \pm 3$:

$$V_{\rm trig} \propto \sum_{k,\sigma,m} c^\dagger_{k,m,\sigma} c_{k,m-3,\sigma} + \text{h.c.} \tag{B3}$$

The auxiliary fermions $O_{k,m,\sigma}$ represent precisely the $m = \pm 3, \pm 6, \ldots$ channels activated by this selection rule.

3. Gaussian Integration and Passive Relative Kernel

Since $H_O + H_{\rm hyb}$ is bilinear in $O_{k,m,\sigma}$, integrating out all $m \neq 0$ modes is a Gaussian path integral within the quadratic auxiliary sector. For fixed (k, θ, σ) the auxiliary action has exactly the form

$$S_O + S_{cO} = \int_{\mathcal{C}} dt dt'\, \bar{O}(t) A_O(t, t') O(t') + \int_{\mathcal{C}} dt\, \left[\bar{c}(t) V_\sigma(t) O(t) + \bar{O}(t) V^\dagger_\sigma(t) c(t) \right]. \tag{B4}$$

The Grassmann integral gives the exact quadratic influence action

$$S_{\rm infl} = -\int_{\mathcal{C}} dt dt'\, \bar{c}(t) V_\sigma(t) g_O(t,t') V_\sigma^\dagger(t') c(t'), \qquad \text{(B5)}$$

so the explicit retarded vertex contraction is

$$\Sigma_\sigma^{O,R}(t,t') = \sum_{m,m'\neq 0} V_{m\sigma}(t)\, g_{mm'}^{O,R}(t,t')\, V_{m'\sigma}^*(t'). \qquad \text{(B6)}$$

No anomalous auxiliary propagator is used. For a number-conserving Hermitian auxiliary Hamiltonian this contraction contains the phase difference fixed by the vertex and its Hermitian conjugate. The Wigner step connecting this two-time expression to the frozen bath self-energy can be displayed explicitly. Set $T = t_{\rm av} = (t+t')/2$ and $\tau = t - t'$, and write $\mathcal{V}_{mm'} = V_m V_{m'}^* e^{i3(m-m')\theta}$. To first adiabatic order,

$$\begin{aligned} &V_{m\sigma}(T+\tau/2) V_{m'\sigma}^*(T-\tau/2) \\ &\quad = \mathcal{V}_{mm'} e^{i\zeta_\sigma[\phi(T+\tau/2)-\phi(T-\tau/2)]} \\ &\quad = \mathcal{V}_{mm'} e^{i\zeta_\sigma \tau \dot{\phi}(T)} \left[1 + O(\tau^3 \partial_T^3 \phi)\right]. \end{aligned} \qquad \text{(B7)}$$

For a stationary auxiliary propagator, its Wigner transform is consequently

$$\begin{aligned} \Sigma_\sigma^{O,R}(\omega, T) &= \Sigma_0^{O,R}[\omega + \zeta_\sigma \dot{\phi}(T)] \\ &= \Sigma_0^{O,R}(\omega) + \zeta_\sigma \dot{\phi}(T) \partial_\omega \Sigma_0^{O,R}(\omega) \\ &\quad + O[(\Omega/\epsilon_O)^2]. \end{aligned} \qquad \text{(B8)}$$

Here the remainder denotes the relative gradient correction in the off-shell window $|\omega - \epsilon_O^{(m)}| \sim \epsilon_O$. Defining

$$\Sigma_0^{\rm bath,R} = \Sigma_{\rm bg}^R + \Sigma_0^{O,R}, \qquad \Delta_{\rm aux} = \dot{\phi}(T) \partial_\omega \,{\rm Re}\, \Sigma_0^{O,R}, \qquad \text{(B9)}$$

therefore gives

$$\begin{aligned} \Sigma_\sigma^{\rm bath,R}(\omega, T) = &\Sigma_0^{\rm bath,R}(\omega) + \zeta_\sigma \Delta_{\rm aux}(\omega, T) \\ &+ O[(\Omega/\epsilon_O)^2], \qquad \Delta_{\rm aux} \in \mathbb{R}. \end{aligned} \qquad \text{(B10)}$$

For diagonal remote modes,

$$\begin{aligned} \Sigma_0^{O,R}(\omega) &= \sum_{m\neq 0} \frac{|V_m|^2}{\omega - \epsilon_O^{(m)} + i0^+}, \\ \Delta_{\rm aux} &= -\dot{\phi}(T) \sum_{m\neq 0} |V_m|^2 \mathcal{P} \frac{1}{[\omega - \epsilon_O^{(m)}]^2} \\ &= -\dot{\phi}(T) \sum_{m\neq 0} |V_m|^2 \partial_{\epsilon_O^{(m)}} \mathcal{P} \frac{1}{\omega - \epsilon_O^{(m)}}. \end{aligned} \qquad \text{(B11)}$$

Thus the off-shell auxiliary sector produces a real principal-value correction in the low-energy window. If an auxiliary pole enters that window, the $-i\pi\delta(\omega - \epsilon_O^{(m)})$ part must instead be retained as ordinary on-shell bath damping. The same frozen shift follows after the unitary rotation $O_{k,m,\sigma} \to e^{-i\zeta_\sigma \phi(t)} O_{k,m,\sigma}$, which removes the drive phase from the vertices and shifts the auxiliary levels by $-\zeta_\sigma \dot{\phi}(t)$. The passive non-Hermitian component appears only after this shifted channel is transmitted through the retarded Dirac propagator. The point is clearest from the quadratic (d, c, O) inverse Green-function kernel,

$$\mathcal{G}^{-1} = \begin{pmatrix} g_d^{-1} & -W^\dagger & 0 \\ -W & g_c^{-1} & -V_\sigma \\ 0 & -V_\sigma^\dagger & g_O^{-1} \end{pmatrix}. \qquad \text{(B12)}$$

Here W is static and all explicit drive dependence resides in $V_\sigma(t)$. Integrating out O and then c gives the two exact Schur complements

$$G_{c,\sigma}^{-1} = g_c^{-1} - V_\sigma g_O V_\sigma^\dagger, \qquad \Sigma_{d,\sigma} = W^\dagger G_{c,\sigma} W. \qquad \text{(B13)}$$

Equivalently,

$$\Sigma_{d,\sigma} = W^\dagger g_c W + W^\dagger g_c (V_\sigma g_O V_\sigma^\dagger) g_c W + \cdots, \qquad \text{(B14)}$$

which explicitly displays the drive-dependent virtual path $d \xrightarrow{W} c \xrightarrow{V_\sigma(t)} O \xrightarrow{V_\sigma^\dagger(t')} c \xrightarrow{W^\dagger} d$. The order of the two Gaussian integrations is immaterial.

For $\Xi_\sigma^R = \Xi_0^R + \zeta_\sigma \delta\Xi^R$, the general linear correction is

$$\delta\Sigma_d^R = W^\dagger G_{c,0}^R\, \delta\Xi^R\, G_{c,0}^R W. \qquad \text{(B15)}$$

It is complex because $G_{c,0}^R$ is a retarded continuum propagator, even when the off-shell $\delta\Xi^R$ is real. In the rigid-shift limit $\delta\Xi^R \simeq \Delta_{\rm aux}$, with weak frequency dependence of the spin-even auxiliary background,

$$\Sigma_{d,\sigma}^R(\omega, t_{\rm av}) \simeq \Sigma_{d,0}^R(\omega - \zeta_\sigma \Delta_{\rm aux}) \simeq \Sigma_{d,0}^R - \zeta_\sigma \Delta_{\rm aux} \partial_\omega \Sigma_{d,0}^R. \qquad \text{(B16)}$$

Writing $\Sigma_{d,0}^R = \Lambda - i\bar{\Gamma}$, the real derivative gives $\Delta_{\rm coh} = -\Delta_{\rm aux} \partial_\omega \Lambda$ in the σ_z channel—it does *not* renormalize the spin-mixing splitting $\Delta_{\rm eff}\sigma_x$ at this order—while the imaginary derivative gives

$$\Gamma_{\rm PT} = \Delta_{\rm aux}\, \partial_\omega \bar{\Gamma}(\omega_0). \qquad \text{(B17)}$$

Equation (B17) is a scalar rigid-shift estimate, not the exact matrix identity for the SOC bath. Writing $a = (\omega^+ - \varepsilon_{k+})^{-1}$, $b = (\omega^+ - \varepsilon_{k-})^{-1}$, $g_0 = (a+b)/2$, and $g_x = (a-b)/2$, one instead has

$$-\partial_\omega g_0 = g_0^2 + g_x^2, \qquad g_b \sigma_z g_b = (g_0^2 - g_x^2)\sigma_z = ab\, \sigma_z. \qquad \text{(B18)}$$

Thus, in the sign convention of Eq. (16),

$$\begin{aligned} \Gamma_{\rm PT} &= \Delta_{\rm aux} \left[\partial_\omega \bar{\Gamma} - 2\, {\rm Im} \sum_k |W_k|^2 g_{x,k}^2 \right] \\ &= \Delta_{\rm aux}\, {\rm Im} \sum_k |W_k|^2 a_k b_k. \end{aligned} \qquad \text{(B19)}$$

An opposite convention for the spin-odd auxiliary shift reverses the full equation. In particular, when the exact positive-energy total width is flat, $\partial_\omega \bar{\Gamma} = 0$, the relative

decay is generated entirely by the SOC g_x^2 term. The full-k calculation below uses the matrix resolvent itself and therefore retains this term; it does not use Eq. (B17) as its operative mechanism. Here ω_0 denotes the low-energy evaluation point of the frozen kernel; at the compensation energy where $\partial_\omega\Lambda$ changes sign (of order $D_{\rm uv}/e$ for the linear Dirac width with a sharp cutoff), the Kramers–Kronig detuning vanishes, $\Delta_{\rm coh}(\omega_0)=0$, while $\partial_\omega\bar\Gamma$ remains finite. For completeness, take the idealized symmetric Dirac width $\bar\Gamma(\epsilon)=g|\epsilon|\Theta(D_{\rm uv}-|\epsilon|)$. Its principal-value Kramers–Kronig partner is

$$\begin{aligned}\Lambda(\omega)&=\frac{1}{\pi}\mathcal{P}\int_{-D_{\rm uv}}^{D_{\rm uv}}d\epsilon\,\frac{\bar\Gamma(\epsilon)}{\omega-\epsilon}\\&=-\frac{g\omega}{\pi}\ln\left|\frac{D_{\rm uv}^2-\omega^2}{\omega^2}\right|.\end{aligned}\tag{B20}$$

Hence, for $|\omega|\ll D_{\rm uv}$,

$$\begin{aligned}\partial_\omega\Lambda&=-\frac{2g}{\pi}\left[\ln\frac{D_{\rm uv}}{|\omega|}-1\right]+O(\omega^2/D_{\rm uv}^2),\\\partial_\omega\bar\Gamma&=g\ \mathrm{sgn}\,\omega.\end{aligned}\tag{B21}$$

This shows explicitly both the compensation scale $|\omega_0|\simeq D_{\rm uv}/e$ in the low-energy form and the logarithmic growth of $|\Delta_{\rm coh}/\Gamma_{\rm PT}|$ near the Dirac node. For the exact sharp-cutoff expression with $D_{\rm uv}=\pi/4$, numerical solution of $\partial_\omega\Lambda=0$ gives $|\omega_c|=0.24608$, rather than the low-energy estimate $D_{\rm uv}/e=0.28893$. A smooth cutoff shifts this number but not the mechanism. The SOC split Dirac bath provides the needed helicity channels and channel-resolved widths

$$\bar\Gamma_\eta(\omega)=\pi\sum_k|W_{k\eta}|^2\delta[\omega-\varepsilon_{c,\eta}(k)].\tag{B22}$$

After subtracting the common damping $-i\bar\Gamma I$, the remaining co-decaying kernel contains the passive-$\mathcal{PT}$ core $\Delta_{\rm eff}\sigma_x+i\Gamma_{\rm PT}\sigma_z$ of Eq. (35) together with the $\mathcal{PT}$-odd Kramers–Kronig detuning $\Delta_{\rm coh}\sigma_z$; exact passive $\mathcal{PT}$ symmetry holds at the compensated point $\Delta_{\rm coh}=0$, and a finite detuning acts as the causal floor of Eq. (37). Indeed,

$$|s|^4=\left(\Delta_{\rm eff}^2+\Delta_{\rm coh}^2-\Gamma_{\rm PT}^2\right)^2+4\Delta_{\rm coh}^2\Gamma_{\rm PT}^2.\tag{B23}$$

Minimizing over the real control $\Delta_{\rm eff}^2\ge0$ gives

$$|s|_{\rm min}=\begin{cases}\sqrt{2|\Delta_{\rm coh}\Gamma_{\rm PT}|}, & |\Delta_{\rm coh}|\le|\Gamma_{\rm PT}|,\\\sqrt{\Delta_{\rm coh}^2+\Gamma_{\rm PT}^2}, & |\Delta_{\rm coh}|>|\Gamma_{\rm PT}|.\end{cases}\tag{B24}$$

The square-root expression quoted for the near-compensated regime therefore has the necessary domain $|\Delta_{\rm coh}|\le|\Gamma_{\rm PT}|$; outside it the minimum occurs at $\Delta_{\rm eff}=0$. This is the sense in which the strictly Hermitian driven model generates the relative non-Hermitian kernel used in the main text. Compensation is local in frequency: for the audited values $\bar\Gamma=0.12$, $\Gamma_{\rm PT}=0.10$, the full sharp-cutoff kernel one common half-width from ω_c gives $|s|_{\rm min}/\Gamma_{\rm PT}=1.0028$ on one side and 0.9004 on the other. Thus the finite-band problem has an order-$\Gamma_{\rm PT}$ avoided- coalescence floor across a linewidth. A structureless scalar flat bath, for which both $\partial_\omega\bar\Gamma=0$ and $g_x=0$, produces only a coherent spin splitting. Flatness of the *total* radial width alone does not remove the passive relative decay: when the helicity channels are nondegenerate, Eq. (B19) leaves the SOC g_x^2 contribution.

The reduction relies on the following conditions:

(i) Impurity at a C_3-invariant site ($\theta_{\rm imp}=0$ mod $2\pi/3$), so the angular projection leaves a single radial bath channel.

(ii) Off-shell auxiliary harmonics, $\epsilon_O^{(m)}\gg D_{\rm uv}$ and $|V|/\epsilon_O^{(m)}\ll1$, so the auxiliary contribution is a controlled principal-value shift. Residual on-shell weight gives ordinary bath damping rather than the relative $\mathcal{PT}$ channel.

(iii) Slow drive on the auxiliary memory time, $\Omega\ll\epsilon_O^{(m)}$, so the correction is organized in powers of Ω/ϵ_O.

(iv) A retarded bath conversion channel: either an energy-dependent scalar width with $\partial_\omega\bar\Gamma\neq0$, as in the cone-patch estimate, or nonzero SOC matrix dressing g_x, as in the exact radial model.

(v) H_O bilinear: no many-body approximation is made in the auxiliary-mode integration.

These conditions do not guarantee reachability of a chosen normalized core. With $x=|V|/\epsilon_O$, phase amplitude $\pi/4$, and benchmark ratio $\Gamma_{\rm PT}/\bar\Gamma=0.10/0.12$, the leading mapping requires

$$\frac{\Omega}{\epsilon_O}\gtrsim\frac{1.061}{x^2}.\tag{B25}$$

For $x\ll1$ this is incompatible with the off-shell/quasistatic hierarchy. Accordingly the derivation fixes the form and symmetry of the passive relative kernel, while the exact benchmark EP remains a normalized constant-core construction rather than a demonstrated microscopic operating point.

4. Stationary rotating drive and full-momentum nonlinear-pole EP

The preceding obstruction concerns the oscillatory quasistatic protocol. A stationary alternative follows from the same microscopic vertex by taking $\phi(t)=\Omega t$. After the unitary transformation of Eq. (7), the remote levels are shifted by the constant $-\zeta_\sigma\Omega$, so the rotated quadratic Hamiltonian is exactly time independent. For

one remote Hermitian level with vertex V_O, its spin-even and spin-odd contributions to the bath inverse are

$$B_{\rm raw}(z) = V_O^2 \frac{z-\epsilon_O}{(z-\epsilon_O)^2-\Omega^2},$$
$$A(z) = -V_O^2 \frac{\Omega}{(z-\epsilon_O)^2-\Omega^2}. \tag{B26}$$

The constant spin-even principal-value shift is absorbed into the bath chemical potential, $B(z) = B_{\rm raw}(z) - B_{\rm raw}(0)$. This subtraction does not affect the spin-odd term or the pole multiplicity.

For $\alpha_{\rm b} = 1$, scalar momentum-independent impurity hybridization W, and radial cutoff K, the physical-sheet impurity self-energy is

$$\Sigma_d^{\rm I}(z) = W^2 \int_0^K \frac{k\,dk}{2\pi} \left[(z-B-k^2)\mathbb{I} - \lambda k\sigma_x - A\sigma_z\right]^{-1}. \tag{B27}$$

The integral is evaluated analytically and independently checked by adaptive quadrature. Every open helicity channel must be continued. If k_j is an on-shell radial root on the real cut, define

$$P_j = \frac{1}{2}\left[\mathbb{I} + \frac{\lambda k_j \sigma_x + A\sigma_z}{z-B-k_j^2}\right],$$
$$\rho_j = \frac{k_j}{2\pi|v_j|}P_j, \qquad v_j = 2k_j + \frac{\lambda^2 k_j}{z-B-k_j^2}. \tag{B28}$$

Analytic continuation of the roots and projectors then gives the outgoing resonance sheet

$$\Sigma_d^{\rm II}(z) = \Sigma_d^{\rm I}(z) - 2\pi i W^2 \sum_{j\in{\rm open}} \rho_j(z). \tag{B29}$$

At the solution below there are two open channels. Continuing only one cut misses the EP.

With F defined in Eq. (41), the nonlinear EP conditions are

$$F(z_{\rm EP}) = 0, \qquad \partial_z F(z_{\rm EP}) = 0, \qquad \partial_z^2 F(z_{\rm EP}) \neq 0, \tag{B30}$$

together with geometric multiplicity one of the inverse kernel. To certify the zero without evaluating a singular eigenbasis, for each point on a closed control loop in (ϵ_d, V_O) we branch-track the nearby critical point z_c satisfying $\partial_z F(z_c) = 0$ and form

$$\mathcal{D}_{\rm loc}(\epsilon_d, V_O) = F(z_c; \epsilon_d, V_O). \tag{B31}$$

Because $\partial_z^2 F$ is nonzero on the loop, the implicit-function theorem keeps z_c on one smooth critical-point branch. Provided $\mathcal{D}_{\rm loc}$ is nonzero on the boundary, its integer

$$W_C = \frac{1}{2\pi}\Delta_C \arg \mathcal{D}_{\rm loc} \tag{B32}$$

is the Brouwer degree of the smooth map $(\epsilon_d, V_O) \mapsto (\mathrm{Re}\,\mathcal{D}_{\rm loc}, \mathrm{Im}\,\mathcal{D}_{\rm loc})$. A nonzero degree forces at least one enclosed zero; regular zeros contribute the sign of the corresponding real 2×2 Jacobian. At every such zero, $F = \partial_z F = 0$; the retained lower bound on $|\partial_z^2 F|$ makes the zero quadratic, while the rank-one inverse-kernel test below excludes a scalar or diabolic degeneracy. Explicit tracking of the two poles supplies the equivalent EP2 monodromy test: one circuit exchanges their labels and their separation has half-integer winding.

For

$$\lambda = 0.05, \quad W = 0.10, \quad K = \pi/4,$$
$$\epsilon_O = 120, \quad \Omega = 60, \quad T_{\rm b} = 0.1, \tag{B33}$$

the solution is

$$z_{\rm EP} = 0.011972834346 - 0.002807216274i,$$
$$V_O = 1.213422009614, \qquad \epsilon_d = 0.015170349783. \tag{B34}$$

The algebraic residuals are

$$|F| = 1.44\times 10^{-21}, \qquad |\partial_z F| = 1.09\times 10^{-15},$$
$$\partial_z^2 F = 1.686448434 - 0.190336728i. \tag{B35}$$

The two singular values of the inverse kernel are 1.207×10^{-3} and 1.22×10^{-18}, establishing geometric multiplicity one. On the reference ellipse $\delta\epsilon_d = 10^{-4}$ and $\delta V_O = 10^{-2}$, $W[\mathcal{D}_{\rm loc}] = +1$; the two poles exchange and their separation has half-winding $+1/2$. The minimum boundary modulus is 2.10×10^{-9}, the largest phase step is $0.0882 < \pi/2$, and $|\partial_z^2 F|$ remains above 1.697. The maximum pole-tracking step is only 0.103 of the local half-separation, and the maximum pole residual is 8.62×10^{-21}. Halving and doubling both ellipse radii leave the winding and pole exchange unchanged.

The causal and microscopic controls are independent of the root solve. The analytic radial integral agrees with direct quadrature to 7.93×10^{-19}. Matching the upper rim of the physical sheet to the continued lower rim gives a two-cut residual 4.02×10^{-9} at $\eta = 10^{-8}$. The physical rate-matrix eigenvalues are positive, 0.00190368 and 0.00309632, so $\bar\Gamma = 0.00250000$ and $\Gamma_{\rm min}/\bar\Gamma = 0.76147$. The driven positive $k = 0$ threshold is $+0.00817959$; the certified pole is $0.00379325 = 1.351|\,\mathrm{Im}\, z_{\rm EP}|$ above it. Thus both open cuts, the threshold proximity, and the lower-branch van Hove structure are already included in the continuation. Finally,

$$\frac{V_O}{\epsilon_O - \Omega} = 0.02022, \qquad \frac{\Omega}{K} = 76.39,$$
$$\frac{\Omega}{\bar\Gamma} = 2.40\times 10^4, \quad \frac{T_{\rm b}}{\bar\Gamma} = 40, \qquad \frac{\epsilon_O - \Omega}{T_{\rm b}} = 600. \tag{B36}$$

Thus the auxiliary sector is off shell and thermally empty, while its exact frequency dependence has been retained rather than expanded in Ω/ϵ_O. The large $\Omega/\bar\Gamma$ ratio would invalidate an oscillatory quasistatic interpretation but is irrelevant to the exactly stationary rotating-frame

Hamiltonian. The certificate itself is retarded and temperature independent; linewidth-resolved equilibrium observables for this witness require $T_{\rm b} \lesssim 10^{-3}$. Figure S8 displays the local winding and pole exchange.

The same equations can be continued in SOC with W held fixed at 0.10 while retuning the real impurity gate and auxiliary vertex. We obtain a regular one-dimensional locus for $0.02 \le \lambda \le 0.1578$. Representative points are

λ	$\mathrm{Re}\, z_{\rm EP}$	$\mathrm{Im}\, z_{\rm EP}$	ϵ_d	V_O
0.05	0.01197283	−0.00280722	0.01517035	1.21342201
0.10	0.02035701	−0.00292855	0.02323413	1.83318448
0.15	0.03265022	−0.00303645	0.03515182	2.41589290

At all three points $\mathcal{D}_{\rm loc}$ has winding +1, the endpoint pole permutation is (12), and the pole-separation half-winding is +1/2. The maximum tracking-step/half-separation ratios are 0.103, 0.100, and 0.119, with no failed pole solves. The corresponding threshold gaps in pole widths are 1.351, 0.577, and 0.076 and the passivity margins are 0.761, 0.614, and 0.515, respectively. Throughout the continued segment $V_O/(\epsilon_O - \Omega) < 0.042$ and the physical rate matrix remains positive. As shown in Fig. S9, the branch approaches the upper open-channel threshold near the high-SOC end while its passivity margin remains positive. This threshold termination is distinct from a passivity failure in a separate normalized sharp-cutoff audit. We do not classify its endpoint and do not infer a two-dimensional EP phase region.

This calculation establishes a microscopic possibility for a nonlinear causal pole EP in the full curved momentum-summed retarded kernel. It does not show that the normalized $\lambda = W = 0.75$ benchmark is a causal EP, and it does not identify the nonlinear tangency condition $F = \partial_z F = 0$ with the nilpotent constant-matrix condition $M^2 = 0$. Consequently no full-momentum Yang–Baxter equation (YBE), RLL, or RTT structure or many-body Jordan amplification is inferred from this certificate.

This completes the microscopic link asserted at the start of the appendix. The key point is that the Gaussian auxiliary projection gives a real spin-odd shift, and the retarded SOC matrix bath converts that shift into a passive relative gain–loss component in the co-decaying frame, together with its Kramers–Kronig companion detuning. The same SOC off-diagonal dressing supplies the $\mathcal{PT}$-even coherent mixing channel through Eq. (32) and, through its g_x^2 contribution in Eq. (B19), the relative decay even when the total positive-energy radial width is flat.

a. Result: 2D bath → 1D effective bath. After the Gaussian integration, the impurity is coupled to only the $m = 0$ bath channel, labeled by radial momentum k alone. For the full dispersions $\varepsilon_{k\pm} = \alpha_{\rm b} k^2 \pm \lambda k$, define $\Delta(\omega) = \sqrt{\lambda^2 + 4\alpha_{\rm b}\omega}$. When both positive-energy radial roots lie inside the cutoff, their exact densities are

$$\begin{aligned} \rho_+(\omega) &= \frac{\Delta - \lambda}{4\pi\alpha_{\rm b}\Delta}, \quad \rho_-(\omega) = \frac{\Delta + \lambda}{4\pi\alpha_{\rm b}\Delta}, \\ \rho_+ + \rho_- &= \frac{1}{2\pi\alpha_{\rm b}}. \end{aligned} \tag{B37}$$

Thus the complete radial bath has a constant positive-energy floor, not a pseudogap. For $-\lambda^2/(4\alpha_{\rm b}) < \omega < 0$, the lower branch instead gives

$$\rho_-(\omega) = \frac{\lambda}{2\pi\alpha_{\rm b}\sqrt{\lambda^2 + 4\alpha_{\rm b}\omega}}, \tag{B38}$$

with a radial van Hove edge at $-\lambda^2/(4\alpha_{\rm b})$. The $|\omega|/\lambda^2$ Dirac law is the small-k cone contribution of one branch (or a cone-patch model), not the density of states (DOS) of the complete quadratic-plus-SOC radial problem.

The Gaussian reduction leaves a single radial ($m = 0$) bath carrying the exact DOS above, hybridizing with the impurity through the channel-preserving vertex of Eq. (B2). After the spin rotation of Sec. IV B this radial bath splits into the two chiral channels $c_{k\pm}$, each coupled diagonally to a single impurity component $\tilde{d}_\pm$. This diagonal bare vertex does not, however, diagonalize the complete scattering problem: the internal $i\Gamma_{\rm PT}$ impurity matrix element mediates conversion between the two asymptotic bath channels. Section IX therefore first uses eigenphases of the full 2×2 unitary scattering matrix only for exact one-particle phase counting. Multiparticle factorization is asserted only after the additional equal-velocity branchwise linearization and folding stated explicitly there.

The same helicity decomposition also makes clear that the channel-odd self-energy is generally complex:

$$\begin{aligned} g_x &= \frac{1}{2}\left[(\omega^+ - \epsilon_{k+})^{-1} - (\omega^+ - \epsilon_{k-})^{-1}\right], \\ \Sigma_x &= \frac{1}{2}(\Sigma_+ - \Sigma_-) = \frac{\Lambda_+ - \Lambda_-}{2} - i\frac{\Gamma_+ - \Gamma_-}{2}. \end{aligned} \tag{B39}$$

The normalized Markov core denotes only the first, coherent term by $\Delta_{\rm eff}$; the full-k causal certificate retains both terms and their frequency dependence.

Appendix C: Finite-U Matrix Block-Wilson Continuation

This section tests the interaction stability of the nonlinear causal pole EP of Sec. B 4. It is separate from the scalar-hybridization Markov-core residue audit in Sec. L. The present calculation retains the SOC-resolved 2×2 physical-bath hybridization, the analytic two-cut continuation, and the exact frequency dependence of the remote auxiliary level. The low-energy physical bath is represented by a finite matrix Wilson chain; the remote level remains in the analytic one-body kernel.

1. Matrix chain and exact finite-chain response

For the $\lambda = 0.05$, $W = 0.10$ witness of Eq. (B33), the positive matrix spectral measure is logarithmically discretized with Wilson parameter $\Lambda = 1.5$ and mapped by block Lanczos to a chain with 2×2 onsite and hopping blocks. We use 12 chain sites for the endpoint center and pointwise certificate, then repeat the complete-basis calculation at $N = 10$ and 12 for a common-contour chain-length test. Each length averages four independent logarithmic discretization shifts. Star-to-chain reconstruction errors lie between 3.5×10^{-15} and 2.3×10^{-14}.

The impurity interaction is

$$H_U = U n_{d\uparrow} n_{d\downarrow}. \tag{C1}$$

Each finite chain is solved in untruncated fixed-charge sectors for its ground state and $N \pm 1$ correction vectors. Writing $q_\alpha = n_{d\bar{\alpha}} d_\alpha$, we use

$$\begin{aligned} G_{\alpha\beta}(z) &= \langle\langle d_\alpha; d_\beta^\dagger \rangle\rangle_z, \\ F_{\alpha\beta}(z) &= \langle\langle q_\alpha; d_\beta^\dagger \rangle\rangle_z. \end{aligned} \tag{C2}$$

The four chain responses are averaged before forming the exact finite-chain equation-of-motion self-energy

$$\Sigma_U(z) = U F(z) G^{-1}(z). \tag{C3}$$

An independent Dyson construction agrees with Eq. (C3) to maximum absolute residual 4.67×10^{-16} and maximum relative residual 1.32×10^{-11}. At $U = 0$, the exact chain response agrees with the direct one-body Green function to 2.02×10^{-14}; anticommutator sum rules are satisfied at machine precision.

The determinant F_U is obtained by adding Eq. (C3) to the same analytic outgoing-sheet one-body kernel used in Eq. (B27). Simultaneously solving $F_U = \partial_z F_U = 0$ while continuing from $U = 0$ gives

U	$\mathrm{Re}\, z_{\mathrm{EP}}$	$\mathrm{Im}\, z_{\mathrm{EP}}$	ϵ_d	W
0	0.01197283	-0.00280722	0.01517035	0.10000000
0.0010	0.01189782	-0.00255800	0.01483973	0.09577738
0.0015	0.01185780	-0.00244927	0.01468704	0.09385894
0.0020	0.01181724	-0.00234964	0.01454259	0.09205708
0.0025	0.01177670	-0.00225815	0.01440607	0.09036305

The $U = 0.0025$ endpoint is reached both by direct initialization and by stepping from $U = 0.0020$. All four discretization shifts retain ground charge two. In the units of the article, $U/D_{\mathrm{uv}} = 3.18 \times 10^{-3}$; with the retuned local width $\bar{\Gamma}_{\mathrm{self}} = W^2/4 = 0.002050$, however, $U/\bar{\Gamma}_{\mathrm{self}} = 1.225$. The endpoint is therefore finite and moderate relative to the local width but remains far from a bandwidth-scale or deep-Kondo strong-coupling limit.

2. Pointwise topological gate

At the interacting center we evaluate a 64-point ellipse with half-widths $\delta\epsilon_d = 10^{-4}$ and $\delta W = 10^{-3}$. At every point the nearby critical point satisfying $\partial_z F_U = 0$ and the two associated poles are independently branch tracked. The resulting gates are

Diagnostic	Result
Local-discriminant winding	-1.00000000000012
Endpoint pole permutation	(12)
Pole-separation half-winding	-0.50000000000005
Minimum boundary modulus	1.5060×10^{-9}
Maximum phase increment	0.27725
Minimum $\lvert\partial_z^2 F_U\rvert$	1.71153
Maximum $\lvert\partial_z F_U\rvert$	1.1150×10^{-15}
Minimum pole separation	8.3767×10^{-5}
Maximum pole residual	2.5452×10^{-21}
Assignment/half-separation ratio	$0.42415 < 1/2$
Maximum anticommutator sum-rule error	3.7682×10^{-15}

Thus the contour is nondegenerate, its phase sampling is resolved, the critical-point branch remains quadratic, and the pole assignment satisfies the explicit separation guard. The winding and pole exchange in Fig. 1 certify one interacting double zero inside the ellipse without diagonalizing at the defective point.

As a discretization-center check, increasing the average from four to eight shifts moves $\mathrm{Re}\, z_{\mathrm{EP}}$, $\mathrm{Im}\, z_{\mathrm{EP}}$, ϵ_d, and W by 0.022%, 0.71%, 0.14%, and 0.33%, respectively. This is a center-convergence check, not an infinite-chain extrapolation.

3. Common-contour chain-length gate

The $N = 10$ and $N = 12$ interacting centers are solved independently rather than inherited from one another. Their maximum absolute coordinate shift is 8.737×10^{-5} and their maximum relative coordinate shift is 3.746×10^{-3}. We then evaluate both chains on the same ellipse, centered on the reoptimized $N = 12$ point, with half-widths $\delta\epsilon_d = 2 \times 10^{-4}$ and $\delta W = 2 \times 10^{-3}$. At 256 contour points the branch-tracked local discriminant D_N gives the complete-basis gates

Diagnostic	$N = 10$	$N = 12$
Winding	-1	-1
Half-winding	$-1/2$	$-1/2$
Pole permutation	(12)	(12)
$\min_C \lvert D_N\rvert$	2.417×10^{-9}	2.897×10^{-9}
Assignment ratio	0.1388	0.1352
Sum-rule error	1.256×10^{-15}	6.280×10^{-15}

Here "assignment" is the maximum branch-assignment step divided by the local half-separation. It remains well below 1/2 for both chains. The pairwise sufficient inequality is

$$\begin{aligned} \sup_C \lvert D_{12} - D_{10}\rvert &= 1.638 \times 10^{-9} \\ &< \min_C \lvert D_{10}\rvert = 2.417 \times 10^{-9}, \end{aligned} \tag{C4}$$

or a conservative Rouché ratio 0.6775. Consequently the two finite chains have the same number of enclosed zeros on this common contour. The 128-point calculation gives the same winding, exchange, and half-winding, with ratio 0.6758; doubling the resolution halves the assignment ratios and changes the Rouché ratio by only 0.25%.

Equation (C4) is a finite-N stability statement. It does not bound the omitted Wilson-chain tail and therefore does not prove an infinite-chain or thermodynamic limit.

4. Kept-space calibration and scope

To quantify what an ordinary truncated run would resolve, we compare kept-space responses with the exact response on the identical four 12-site chains:

N_{keep}	max. G	max. F	max. Σ_U	sum-rule deficit
320	11.67%	47.35%	38.00%	2.61%
640	11.76%	32.90%	24.74%	1.51%
960	3.71%	4.57%	4.18%	1.40%

The retained deviations decrease at $N_{\mathrm{keep}} = 960$, but this comparison is not used to certify the topology: the load-bearing result is the untruncated complete-basis response on the specified finite chains.

Two limitations are explicit. First, the $N = 10 \to 12$ Rouché pass is a controlled comparison of two complete-basis finite chains, but chain-length and Λ convergence to the continuum still require a true complete-basis spectral numerical renormalization group (NRG) or density-matrix NRG (DM-NRG) implementation; at 12 sites the omitted hopping is not parametrically below the pole scale. Second, a short-chain four-band discretization including the remote auxiliary pair selects ground charges $1, 2, 2, 1$ across the four shifts. Its center is contaminated by finite-size charge steps and is reported neither as a pass nor as a failure. The certified calculation therefore establishes a zero-temperature interacting EP of the finite low-energy matrix chains dressed by the exact analytic auxiliary kernel. It does not establish a thermodynamic full-density-matrix NRG (FDM-NRG) fixed point, finite-temperature observability, a phase boundary, or universal interaction scaling.

Appendix D: Keldysh Mean-Field Theory

The slave-boson representation $d^\dagger_\sigma = f^\dagger_\sigma b$ introduces spinon fermions f_σ (the Kondo quasiparticles) and a slave boson b (enforcing no double occupancy) with condensate $b_c = \langle b \rangle$ and Lagrange multiplier δ_c enforcing the constraint $r^2 + \sum_\sigma f^\dagger_\sigma f_\sigma = Q$ (where $r = |b_c|$). Here $Q = 1$ is the physical spin-1/2 constraint, while $Q = N q_{\mathrm{occ}}$ denotes the large-N or channel-rescaled normalization used in the corresponding numerical saddle. The quadratic spinon action can be organized either with an explicit bath or with its complete influence kernel. In the latter representation the retarded impurity kernel has the schematic form

$$\mathcal{G}^{R,-1}_{ff} = \omega - \epsilon_d - \delta_c - r^2 D_0(\beta_0)\sigma_x - r^2 K^R_\Gamma(\omega, \beta_0), \quad \text{(D1)}$$

and its lesser partner contains $r^2 K^<_\Gamma$. Therefore

$$\begin{aligned} \partial_r[rW] = W, \qquad \partial_r[r^2 D_0] = 2r D_0, \\ \partial_r[r^2 K_\Gamma] = 2r K_\Gamma. \end{aligned} \quad \text{(D2)}$$

These three derivatives respectively generate the hybridization, coherent-channel, and dissipative-influence forces below. For the $U \to \infty$ Coleman saddle, variation of the complete quadratic contour action gives the basis-independent equations

$$0 = r^2 + n_f - Q, \qquad n_f = -i \int \frac{d\omega}{2\pi} \mathrm{Tr}_s G^<_{ff}(\omega), \quad \text{(D3)}$$

$$0 = 2r\delta_c + \mathcal{F}_W + 2r D_0\, \mathcal{X} + 2r \mathcal{I}_\Gamma. \quad \text{(D4)}$$

Here

$$\mathcal{F}_W = -i \int \frac{d\omega}{2\pi} \sum_k \mathrm{Tr}_s \left[V_H G^<_{cf,k} + V^\dagger_H G^<_{fc,k} \right], \quad \text{(D5)}$$

$$\mathcal{X} = -i \int \frac{d\omega}{2\pi} \mathrm{Tr}_s [\sigma_x G^<_{ff}], \quad \text{(D6)}$$

$$\mathcal{I}_\Gamma = \mathrm{Re} \left\{ -i \int \frac{d\omega}{2\pi} \mathrm{Tr}_s \left[K^R_\Gamma G^<_{ff} + K^<_\Gamma G^A_{ff} \right] \right\}. \quad \text{(D7)}$$

The last line is the Langreth projection of the derivative of the dissipative influence action. Dropping $K^<_\Gamma G^A$ generally produces a complex force and is not a valid thermal saddle. If the relevant bath is retained explicitly instead, $\mathcal{I}_\Gamma$ is absent as a separate term because it is already contained in $\mathcal{F}_W$; the two descriptions must not be combined. Equation (D4) is divided by $2r$ only after imposing a finite-r guard. At the $r = 0$ boundary the undivided equation is used, so the numerical update never introduces a spurious $1/r$ singularity. Under the Hadamard rotation, $\sigma_x \mapsto \sigma_z$, $\sigma_z \mapsto \sigma_x$, and $V_H \mapsto W_k I$. Hence in the chiral basis

$$\mathcal{F}^\pm_W = -i \int \frac{d\omega}{2\pi} \sum_{k,\eta} W_k \left(G^<_{c_\eta f_\eta, k} + G^<_{f_\eta c_\eta, k} \right), \quad \text{(D8)}$$

$$\mathcal{X}_\pm = -i \int \frac{d\omega}{2\pi} (G^<_{++} - G^<_{--}), \quad \text{(D9)}$$

while the relative dissipative force depends on the off-diagonal chiral components because $\widetilde{K}_\Gamma \propto iG\sigma_x$. The full matrix lesser Green function must therefore be retained. The lesser Green's functions are evaluated from the bath sandwich $G^< = G^R \Sigma^<_{\mathrm{bath}} G^A$, with $\Sigma^<_{\mathrm{bath}} = if\Gamma_{\mathrm{bath}}$. When the complete thermal influence kernel is retained this is algebraically equivalent to $G^< = f(\omega)[G^A - G^R]$ [1]; the sandwich form is used in the solver so that the channel matrix and charge-continuity test remain explicit. This is an instantaneous-equilibrium

closure at leading adiabatic order. Corrections of order $O[(\Omega/\epsilon_O)^2]$ and the corresponding Wigner gradients would encode drive-induced redistribution; no energy-pumping result is asserted at the retained order. For the distinct stationary rotating-drive witness, no Wigner freezing is used. The impurity and physical bath fields are untransformed, while the nearest shifted auxiliary level lies $(\epsilon_O-\Omega)/T_\mathrm{b} = 600$ above the chemical potential. Its lesser occupation is therefore bounded by e^{-600} within the discrete off-shell closure; the thermal bath lesser kernel used for that retarded witness is not an adiabatic assumption. The causal constraint on the Lagrange multiplier is implemented by projecting $\delta_c \leftarrow \mathrm{Re}(\delta_c)$ at each iteration, which is equivalent to enforcing the retarded (causal) boundary condition.

a. Scope of the saddle equations. Equations (D3)–(D7) specify the consistent $U \to \infty$ saddle. The finite-$U = 2$ two-denominator Schrieffer–Wolff expressions used later are a separate frozen analytic estimate and are not inserted into this saddle. The principal spectral figures use frozen r and δ_c and therefore do not claim a finite-temperature self-consistent enhancement of the condensate. The phase of b_c is a $U(1)$ gauge choice throughout.

The physical projected channels are those of Eq. (A3), with $V_\mathrm{eff} = rW$. Main-text Fig. 2 sets $r = 1$ and uses a normalized geometry scan to display the bifurcation. Supplemental Fig. S5 instead uses fixed $\Gamma_0 = \pi\rho W^2 = 0.12$ and $\Delta_\mathrm{eff}^{(0)} = 0.10$, $\Gamma_\mathrm{PT}^{(0)} = 0.20\beta_0$. Hence W is independent of β_0 in the quantitative small-width validation.

1. Low-temperature self-consistent gate

The corrected $U \to \infty$ saddle was solved for the fixed-W small-width path with the complete thermal Markov influence kernel. At $T_\mathrm{b} = 10^{-6}$, all real, negative-real, and complex initial boson seeds converge to the same gauge-fixed solution over $0.35 \le \beta_0 \le 0.55$. The amplitude decreases smoothly from $r = 0.0106078$ to 0.00949995; at the core point $\beta_0 = 0.50$, $r = 0.00980741$ and $\delta_c = 1.00000012$. Across the scan the maximum constraint residual is 6.54×10^{-8}, the maximum divided stationarity residual is 4.86×10^{-10}, the maximum grid-refinement error is 1.71×10^{-6}, and the charge-continuity residual is 3.66×10^{-23}.

Because both $D = r^2 D_0$ and $G = r^2 G_0$, the local core condition remains $D_0^2 = G_0^2$ and its root is exactly $\beta_0 = 0.50$ for every finite r. At $k = 0$ the two SOC bath energies coincide, so the Schur dressing is a common scalar shift and the dressed double-root condition is also exactly the core condition. For every tested nonzero k, the minimizing β_0 lies at the upper boundary 0.55 of the scan; no interior finite-k EP is therefore inferred.

At the reference temperature $T_\mathrm{b} = 0.1$, the same closure has no finite-r solution on this parameter path and reaches the $r = 0$ boundary. This is why the $T_\mathrm{b} = 0.1$ spectra remain frozen diagnostics and why no finite-temperature condensate or Kondo-scale enhancement is inferred. Figure S2 reports both the converged low-temperature branch and the numerical validation checks.

a. Scope of Fig. 2. Figure 2 is a normalized $k = 0$ constant-core calculations with fixed $W = 0.75$; the SOC term vanishes at that momentum, and no physical DOS, cutoff, temperature, or interaction parameter enters those panels. The exact finite-U statement belongs instead to the controlled linearized contact model of Sec. H 4; there the standard local interaction preserves the symmetry-protected EP.

Appendix E: Bath-Controlled FDR and Physical Spectral Function

The complete reservoir-coupled problem uses the Keldysh lesser Green's function fixed by the thermal bath. No independent mode-by-mode occupation prescription enters the physical occupations or saddle-point equations.

a. Physical bath construction. With $G^<_{ij}(t,t') = i\langle d_j^\dagger(t') d_i(t)\rangle$ and hence $n_f = -i\,\mathrm{Tr}\, G^<(t,t)$,

$$\begin{aligned}\Sigma^<_\mathrm{bath} &= i f \Gamma_\mathrm{bath},\\ G^<_\mathrm{bath} &= G^R \Sigma^<_\mathrm{bath} G^A = f(G^A - G^R)\end{aligned} \tag{E1}$$

follows directly from the Keldysh identity for a system bilinearly coupled to a thermal reservoir [1]. It satisfies the bath-controlled FDR for this bilinear reservoir setup, independent of whether H is Hermitian. In the driven problem this is the leading instantaneous adiabatic closure; finite-frequency pumping requires the omitted higher-gradient corrections. The need for a complete doubled-contour construction is consistent with the exact high-bias Anderson-impurity solution of Oguri and Sakano [2], where a finite non-Hermitian Liouville–Fock Hamiltonian reproduces physical Keldysh Green functions only with the appropriate paired initial and final states. Their high-energy limit is cited as a structural precedent, not as a low-temperature approximation to the present driven kernel. In the diagonal-hybridization basis of $\tilde{H}$, the two channels $\tilde{d}_\pm$ are independently coupled to the $c_{k\pm}$ baths, and each satisfies the FDR: $[G^<_\mathrm{bath}]_{\pm\pm} = f(\omega)[G^A - G^R]_{\pm\pm}$.

b. Number continuity. The same complete self-energy satisfies the stationary continuity identity

$$\mathcal{I}_N = \int \frac{d\omega}{2\pi}\,\mathrm{Tr}\big[\Sigma^< G^> - \Sigma^> G^<\big] = 0. \tag{E2}$$

This check uses the full channel matrix, including the thermal partner of the relative retarded width, and independently validates the bath construction.

c. Physical DOS identity. For the complete physical kernel, $G^A = (G^R)^\dagger$ and

$$\mathcal{A} = i(G^R - G^A) = G^R \Gamma_\mathrm{tot} G^A, \qquad \Gamma_\mathrm{tot} \succeq 0. \tag{E3}$$

Consequently,

$$\rho_{\rm imp} = \frac{1}{2\pi}\mathrm{Tr}(P_{\rm imp}\mathcal{A}) = -\frac{1}{\pi}\mathrm{Im}\,\mathrm{Tr}(P_{\rm imp}G^R). \quad \text{(E4)}$$

The two expressions are identical, not competing density-of-states (DOS) prescriptions. We verify their equality and the positivity of the complete physical response numerically. A signed response of the compensated relative kernel after common damping is removed is only a kernel diagnostic and is not interpreted as a physical density of states.

Appendix F: Shifted Friedel-Type Resonance Condition

This appendix gives the conservative form of the shifted resonance condition used in Sec. V D. The purpose is not to claim a new exact sum rule, but to identify the reference position of the low-energy peak within the SBMF Green's function.

1. Resonance Position in the Lorentzian Regime

The spinon retarded Green's function at the SBMF saddle point may be written as

$$G^R_{f,\sigma}(\omega) = \frac{1}{\omega - \tilde{\epsilon}_\xi + i\Gamma^{\rm net}_\sigma}, \qquad \Gamma^{\rm net}_\sigma = \Gamma_\sigma - x_\sigma, \quad \text{(F1)}$$

where $x_\sigma = \zeta_\sigma\Gamma_{\rm PT} = \pm\Gamma_{\rm PT}$ is the spin-selective imaginary shift from the coarse-grained drive and $\Gamma^{\rm net}_\sigma > 0$ in the stable low-energy regime. The corresponding physical impurity spectral function is

$$A_{\rm imp}(\omega) = \frac{1}{\pi}\sum_\sigma \frac{\Gamma^{\rm net}_\sigma}{(\omega - \mathrm{Re}\,\tilde{\epsilon}_\xi)^2 + (\Gamma^{\rm net}_\sigma)^2}. \quad \text{(F2)}$$

When the line shape is approximately Lorentzian, both spin channels are centred at the same reference frequency

$$\omega_{\rm res} \simeq \mathrm{Re}(\tilde{\epsilon}_\xi). \quad \text{(F3)}$$

This is the shifted Friedel-type spectral resonance condition used in the figures. In the strongly non-Lorentzian regime, the observed maximum can deviate from this reference; those deviations are treated as line-shape effects rather than as a breakdown of a sum rule.

2. Origin of the Shift

In the Hermitian particle-hole-symmetric SBMF problem, $\epsilon_\xi = -U/2$ and the spin occupations are equal, so the saddle point pins $\mathrm{Re}(\tilde{\epsilon}_\xi)$ near zero. In the driven problem the coarse-grained self-energy gives unequal effective widths $\Gamma^{\rm net}_\uparrow \neq \Gamma^{\rm net}_\downarrow$ and a real projected shift δ_Σ, which together move the saddle value through

$$\mathrm{Re}(\tilde{\epsilon}_\xi) = \epsilon_\xi + \mathrm{Re}(\delta_c) + \delta_\Sigma. \quad \text{(F4)}$$

The shift is therefore a gauge-invariant consequence of the spin-selective self-energy and of the SBMF saddle point. It is not attributed to the phase of b_c.

a. Numerical reference. For the frozen channel-projected kernel used in the figures, $\mathrm{Re}(\tilde{\epsilon}_\xi)$ is the renormalized impurity level generated by the same plotted data pipeline. In the local-moment reference with $\epsilon_\xi = -U/2 = -1.0$, the frozen closure gives $\mathrm{Re}(\tilde{\epsilon}_\xi) \simeq -1.0$. If a separate finite-temperature self-consistent SBMF closure is used, Eq. (F4) should instead be evaluated with that closure and the figure should be regenerated from the same saddle.

Appendix G: Slave-Boson Mean-Field Kondo Scales

This appendix summarizes the slave-boson mean-field (SBMF) scale estimate in the rotated spinon basis. The emphasis is on the spin-selective passive relative decay shifts, the channel-resolved resonance width, and the way an EP-active Jordan residue can modify the screening kernel without implying a universal exact Kondo temperature.

1. Effective Spinon Hamiltonian

Using the slave-boson representation $d^\dagger_\sigma = f^\dagger_\sigma b$ with gauge-invariant condensate amplitude $r = |b_c|$, the quadratic effective Hamiltonian for the spinon fermions f_σ is:

$$H_{\rm eff} = \sum_\sigma \tilde{\epsilon}_\sigma f^\dagger_\sigma f_\sigma + \sum_{k\sigma}\big[rV_{k\sigma}\, c^\dagger_{k\sigma} f_\sigma + \mathrm{h.c.}\big], \quad \text{(G1)}$$

where $V_{k\sigma}$ is the hybridization and $\tilde{\epsilon}_\sigma = \tilde{\epsilon}_\xi + ix_\sigma$ is the complex spinon level. The chiral spinon modes $\tilde{f}_\pm = (f_\uparrow \pm f_\downarrow)/\sqrt{2}$ are the same rotated basis used in App. D. The spin-dependent projected imaginary shift $x_\sigma \equiv \zeta_\sigma\Gamma_{\rm PT} = \pm\Gamma_{\rm PT}$ (with $\zeta_\uparrow = +1$, $\zeta_\downarrow = -1$) encodes the drive-induced passive relative gain/loss; the real part $\epsilon_{r,\sigma} = \mathrm{Re}(\tilde{\epsilon}_\sigma) = \tilde{\epsilon}_\xi$ is the renormalized level common to both spins. The renormalized hybridization width is:

$$\Gamma_\sigma(\omega) = \pi r^2 \sum_k |V_{k\sigma}|^2\delta(\omega - \varepsilon_{k\sigma}) \equiv r^2\Gamma^{(0)}_\sigma(\omega), \quad \text{(G2)}$$

and the spinon retarded Green's function is:

$$G^R_{f,\sigma}(\omega) = \frac{1}{\omega - \tilde{\epsilon}_\sigma + i\Gamma_\sigma}. \quad \text{(G3)}$$

2. Saddle-Point Equations at $T_{\rm b} = 0$

The constraint $r^2+\sum_\sigma\langle f_\sigma^\dagger f_\sigma\rangle = Q$ and the stationarity condition $\partial\mathcal{F}_{\rm MF}/\partial\delta_c = 0$ give at $T_{\rm b} = 0$:

$$Q - r^2 = \sum_\sigma \int_{-D_{\rm uv}}^{0} \frac{d\omega}{\pi} \frac{\Gamma_\sigma}{(\omega-\epsilon_{r,\sigma})^2+\Gamma_\sigma^2}, \tag{G4}$$

$$\delta_c = \sum_\sigma \int_{-D_{\rm uv}}^{0} \frac{d\omega}{\pi} \frac{(\omega-\epsilon_{r,\sigma})\Gamma_\sigma}{(\omega-\epsilon_{r,\sigma})^2+\Gamma_\sigma^2}. \tag{G5}$$

These are the standard SBMF equations evaluated with the complex spinon level $\tilde\epsilon_\sigma = \epsilon_{r,\sigma} + ix_\sigma$.

3. Kondo-scale estimate near an EP

Within slave-boson mean-field theory the low-energy scale is set by the renormalized impurity width. With $r = |b_c|$ and $V_{\rm eff} = rV$,

$$\tilde\Gamma_\sigma(0) = r^2\Gamma_\sigma^{(0)}(0), \qquad \Gamma_\sigma^{(0)}(0) = \pi V^2\rho_{b,\sigma}(0). \tag{G6}$$

Matching to the usual poor-man's-scaling form gives the channel estimate

$$T_{K,\sigma} \simeq D_{\rm uv}\exp\left[-\frac{\pi|\tilde\epsilon_\sigma|}{2\Gamma_\sigma^{(0)}(0)}\right], \qquad |\tilde\epsilon_\sigma| = \sqrt{\epsilon_{r,\sigma}^2 + x_\sigma^2}, \tag{G7}$$

where $x_\sigma = \zeta_\sigma\Gamma_{\rm PT}$ is the projected spin-selective imaginary shift. The sign difference $x_\uparrow = -x_\downarrow$ is essential for the $\mathcal{PT}$ structure, but it does not by itself split T_K in Eq. (G7), because only x_σ^2 enters. Distinct channel scales arise when the SOC-split bath gives $\Gamma_\uparrow^{(0)}(0) \neq \Gamma_\downarrow^{(0)}(0)$, or when the real detunings $\epsilon_{r,\sigma}$ differ.

Let $h = \Delta_{\rm eff}\sigma_x + i\Gamma_{\rm PT}\sigma_z$, $h^2 = s^2 I$, and $a(z) = z - \epsilon_0 - \Sigma_0^R(z)$. Away from the EP, the biorthogonal projectors and complete resolvent are

$$P_\pm = \frac{1}{2}\left(I \pm \frac{h}{s}\right), \tag{G8}$$

$$\frac{P_+}{a-s} + \frac{P_-}{a+s} = \frac{aI+h}{a^2-s^2}. \tag{G9}$$

Although $P_\pm$ separately grow as $1/s$, their singular pieces cancel in the sum. At the EP, $h_{\rm EP}^2 = 0$ and the limit is

$$(aI - h_{\rm EP})^{-1} = \frac{I}{a} + \frac{h_{\rm EP}}{a^2}. \tag{G10}$$

The double pole produces a $te^{-iE_{\rm EP}t}$ transient; it does not change contour time ordering or supply a free multiplicative coupling.

The same cancellation occurs in the finite-U virtual charge sectors. At particle–hole symmetry with $A = U/2$,

$$(AI-h)^{-1} + (AI+h)^{-1} = \frac{2A}{A^2-s^2}I, \tag{G11}$$

$$J_{\rm SW}(s) = \frac{4AW^2}{A^2-s^2}I,$$

$$J_{\rm SW}(0) = \frac{8W^2}{U}. \tag{G12}$$

Thus no $1/s$ factor remains in the exchange. Under the reciprocal eigenvector rescaling $|r_\nu\rangle \to a_\nu|r_\nu\rangle$ and $\langle l_\nu| \to \langle l_\nu|/a_\nu$, both P_ν and every consistently transformed vertex–propagator–vertex product are invariant. Accordingly, no Petermann, condition-number, or $Z^{\rm bio}$ multiplier is attached to $G^{R,A,<}$, a diagrammatic vertex, or the Kondo exponent. For example, a Gaussian physical response is assembled from the full matrix propagators,

$$\chi_{AB}^R(\omega) = -i\int\frac{d\nu}{2\pi}\big\{\mathrm{Tr}[AG^R(\nu+\omega)BG^<(\nu)] + \mathrm{Tr}[AG^<(\nu+\omega)BG^A(\nu)]\big\}, \tag{G13}$$

so all eigenbranches, projectors, and consistently transformed vertices are summed before a physical contraction is taken. Equation (G13) illustrates the cancellation but is not needed to prove it: the proof is the resolvent identity in Eq. (G9). Conversely, that identity does not remove genuine dependence carried by pole motion, matrix residues, interaction vertices, or relaxation rates. The finite-U matrix-chain and transition-residue calculations below are independent physical-basis and sum-rule consistency tests; they are not presented as a second algebraic proof.

If $\Delta_{\rm eff} = D_0$ and $\Gamma_{\rm PT} = g_G\beta_0$, then

$$\partial_{\beta_0} s^2 = -2\Gamma_{\rm PT} g_G. \tag{G14}$$

Both amplitudes can increase while s either increases or decreases. When s^2 increases, Eq. (G12) can increase through the ordinary charge denominators; this is not geometric amplification. For the pre-EP path used in the direct RG test, s^2 decreases toward zero. The complete causal matrix flow gives active/control scale ratios 1.0567, 1.0179, 0.9893, and 0.9166 at thresholds $g_* = 0.30$, 0.40, 0.50, and 1.00, respectively. This verifies a bounded early-flow ordering but not a threshold-robust Kondo-scale enhancement.

Appendix H: Full Scattering Matrix and Contact Algebra

The Fisher–Lee and Wigner–Smith constructions used here are standard scattering tools [3–5]. Their role is to test causality and phase counting for the projected kernel; the novelty claimed in the article is the microscopic origin of that kernel.

1. Spin, chiral, and scattering-eigenchannel bases

Let $\Psi_s = (d_\uparrow, d_\downarrow)^T$ denote the original spin basis and $U_H = 2^{-1/2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. For the compensated Markovian core,

$$\begin{aligned} h_s &= \tilde{\epsilon}_\xi I + D\sigma_x + iG\sigma_z - i\bar{\Gamma} I, \\ \Gamma_s &= i(\Sigma_s^R - \Sigma_s^A) = 2(\bar{\Gamma} I - G\sigma_z), \end{aligned} \tag{H1}$$

where $D = \Delta_{\rm eff}$ and $G = \Gamma_{\rm PT}$. With $z = \omega - \tilde{\epsilon}_\xi + i\bar{\Gamma}$ and $\mathcal{D}(z) = z^2 - D^2 + G^2$, direct inversion gives

$$G_s^R(\omega) = \frac{1}{\mathcal{D}(z)} \begin{pmatrix} z + iG & D \\ D & z - iG \end{pmatrix}. \tag{H2}$$

Writing $a = z - iG$ and $b = z + iG$ also gives the Schur forms

$$\begin{aligned} [G_s^R]_{\uparrow\uparrow} &= \frac{1}{a - D^2/b}, \qquad [G_s^R]_{\downarrow\downarrow} = \frac{1}{b - D^2/a}, \\ [G_s^R]_{\uparrow\downarrow} &= [G_s^R]_{\downarrow\uparrow} = \frac{D}{ab - D^2}. \end{aligned} \tag{H3}$$

Thus the local denominators are useful Schur complements, but they do not make the complete Green function or scattering matrix diagonal.

The fixed Hadamard/chiral fields are $\Psi_\chi = U_H^\dagger \Psi_s = (\widetilde{d}_+, \widetilde{d}_-)^T$. Every object, including the rate matrix, must be rotated:

$$\begin{aligned} h_\chi &= U_H^\dagger h_s U_H = \tilde{\epsilon}_\xi I + D\sigma_z + iG\sigma_x - i\bar{\Gamma} I, \\ \Gamma_\chi &= U_H^\dagger \Gamma_s U_H = 2(\bar{\Gamma} I - G\sigma_x), \\ G_\chi^R &= U_H^\dagger G_s^R U_H = \frac{1}{\mathcal{D}(z)} \begin{pmatrix} z + D & iG \\ iG & z - D \end{pmatrix}. \end{aligned} \tag{H4}$$

If $p = z - D$ and $q = z + D$, the corresponding chiral Schur coefficients are

$$\begin{aligned} [G_\chi^R]_{++} &= \frac{1}{p + G^2/q}, \qquad [G_\chi^R]_{--} = \frac{1}{q + G^2/p}, \\ [G_\chi^R]_{+-} &= [G_\chi^R]_{-+} = \frac{iG}{pq + G^2}. \end{aligned} \tag{H5}$$

Equations (H3) and (H5) are the corrected coefficient evaluation replacing the former block-diagonal contact argument.

2. Contact matching and Fisher–Lee coefficients

Choose an asymptotic-channel coupling matrix $\mathcal{W}_\nu$ such that $2\pi \mathcal{W}_\nu \mathcal{W}_\nu^\dagger = \Gamma_\nu$, with $\nu = s, \chi$. For an incoming channel vector $A_\nu^{\rm in}$, the impurity contact amplitude and outgoing vector satisfy

$$\begin{aligned} e_\nu(\omega) &= G_\nu^R(\omega) \mathcal{W}_\nu A_\nu^{\rm in}, \\ A_\nu^{\rm out} &= A_\nu^{\rm in} - 2\pi i \mathcal{W}_\nu^\dagger e_\nu = S_\nu(\omega) A_\nu^{\rm in}. \end{aligned} \tag{H6}$$

For the positive square-root factorization this gives

$$S_\nu = I - i\Gamma_\nu^{1/2} G_\nu^R \Gamma_\nu^{1/2}, \qquad S_\chi = U_H^\dagger S_s U_H. \tag{H7}$$

In the spin representation define $\gamma_\uparrow = 2(\bar{\Gamma} - G)$ and $\gamma_\downarrow = 2(\bar{\Gamma} + G)$. The explicit coefficients are

$$\begin{aligned} [S_s]_{\uparrow\uparrow} &= 1 - i\gamma_\uparrow \frac{z + iG}{\mathcal{D}(z)}, \\ [S_s]_{\downarrow\downarrow} &= 1 - i\gamma_\downarrow \frac{z - iG}{\mathcal{D}(z)}, \\ [S_s]_{\uparrow\downarrow} &= [S_s]_{\downarrow\uparrow} = -i\sqrt{\gamma_\uparrow \gamma_\downarrow} \frac{D}{\mathcal{D}(z)}. \end{aligned} \tag{H8}$$

The chiral coefficients must be obtained from the same covariant rotation, or equivalently from Eq. (H7) using Γ_χ and G_χ^R. It is incorrect to rotate G^R while leaving Γ diagonal.

For the validation point $\bar{\Gamma} = 0.12$ and $D = G = 0.10$, at $\omega = \tilde{\epsilon}_\xi$,

$$\begin{aligned} S_s &= \begin{pmatrix} \mathsf{a} & i\mathsf{b} \\ i\mathsf{b} & \mathsf{a} \end{pmatrix}, \\ S_\chi &= \mathrm{diag}(\mathsf{a} + i\mathsf{b}, \mathsf{a} - i\mathsf{b}), \\ \mathsf{a} &= 0.3888889, \qquad \mathsf{b} = 0.9212847. \end{aligned} \tag{H9}$$

Consequently the earlier number 0.9213 is a spin-representation off-diagonal coefficient and, at the resonance center, the imaginary part of the two unit-modulus scattering eigenvalues. It is not a chiral conversion amplitude. In the same Markov gate, $[S_\chi(\tilde{\epsilon}_\xi)]_{+-} = 0$, while $\max_\omega |[S_\chi]_{+-}| = 0.8333333$ at $|\omega - \tilde{\epsilon}_\xi| = 0.12$. These component values are basis dependent; the eigenphases, $\det S$, and Wigner–Smith eigenvalues and trace are the quantities used in the basis-independent checks.

3. Four contact amplitudes and exact folded identity

The four coefficients that occur when the two incoming channels are treated simultaneously are most cleanly organized as the two columns of the impurity contact Green function. Let $e_{\eta'\eta}$ denote the impurity amplitude in fixed chiral component η' generated by an incoming asymptotic channel η. With

$$p = z - D, \qquad q = z + D, \qquad \mathcal{P}_0 = pq + G^2, \tag{H10}$$

the exact quadratic contact equations are

$$\begin{pmatrix} p & -iG & 0 & 0 \\ -iG & q & 0 & 0 \\ 0 & 0 & q & -iG \\ 0 & 0 & -iG & p \end{pmatrix} \begin{pmatrix} e_{++} \\ e_{-+} \\ e_{--} \\ e_{+-} \end{pmatrix} = \mathcal{W} \begin{pmatrix} \Phi_+ \\ 0 \\ \Phi_- \\ 0 \end{pmatrix}. \tag{H11}$$

Here $\mathcal{W}$ is the contact coupling and Φ_η is the incoming amplitude. Equation (H11) is simply the two-column form of Eq. (H4); it is therefore exact for the quadratic

projected kernel and introduces no two-particle assumption.

Rows 2 and 4 give

$$e_{-+} = \frac{iG}{q} e_{++}, \qquad e_{+-} = \frac{iG}{p} e_{--}. \tag{H12}$$

Eliminating them leaves the two direct amplitudes with dressed contact denominators

$$\begin{aligned} \widehat{p}\, e_{++} &= \mathcal{W}\Phi_+, & \widehat{q}\, e_{--} &= \mathcal{W}\Phi_-, \\ \widehat{p} &= p + \frac{G^2}{q} = \frac{\mathcal{P}_0}{q}, & \widehat{q} &= q + \frac{G^2}{p} = \frac{\mathcal{P}_0}{p}. \end{aligned} \tag{H13}$$

Consequently,

$$\frac{\widehat{p}}{\widehat{q}} = \frac{p}{q}. \tag{H14}$$

This identity is useful but must be interpreted correctly. Both folded denominators contain the same quadratic pole polynomial $\mathcal{P}_0$; the ratio removes that common factor. It therefore exposes the common denominator responsible for the Jordan limit and for cancellation between individual projectors. It is not a basis-independent amplification factor. For simultaneous equal sources one obtains $e_{--}/e_{++} = p/q$, whereas changing the incoming superposition changes that amplitude ratio.

4. Controlled finite-U Anderson contact algebra

The four amplitudes in Eq. (H11) are the two columns of a *one-particle* Green function. They should not be confused with the two-electron exchange matrix. To obtain the latter, first impose the controlled reduction stated in the main text: equal folded velocities, linear dispersion, constant $M = D\sigma_z + iG\sigma_x$, momentum-independent scalar hybridization V, and the standard local interaction $Un_{d+}n_{d-}$. With momenta measured in energy units, the two-electron bath wave function in the ordered sector $x_1 < x_2$ is

$$\begin{aligned} \Psi_{ab}(x_1, x_2) = {} & \mathcal{A}_{12}^{ab} e^{ipx_1 + iqx_2} \\ & + \mathcal{A}_{21}^{ab} e^{iqx_1 + ipx_2}, \qquad x_1 < x_2, \end{aligned} \tag{H15}$$

supplemented by sectors with one bath electron and one impurity electron and by a doubly occupied impurity amplitude D_{pq}. Substitution in the Schrödinger equation, integration across the point contact, and elimination of those impurity amplitudes separate into triplet and singlet sectors. In the normalization $\Gamma_A = V^2/(2v)$, the resulting contact equations are [6–9]

$$\mathcal{A}_{21}^{(t)} = \mathcal{A}_{12}^{(t)}, \tag{H16}$$

$$[\Delta\mathcal{B} + i\,2U\Gamma_A]\,\mathcal{A}_{21}^{(s)} = [\Delta\mathcal{B} - i\,2U\Gamma_A]\,\mathcal{A}_{12}^{(s)}, \tag{H17}$$

$$\begin{aligned} \Delta\mathcal{B} &= \mathcal{B}(p) - \mathcal{B}(q), \\ \mathcal{B}(p) &= p(p - 2\epsilon_d - U). \end{aligned} \tag{H18}$$

Equations (H16) and (H17) display exactly where the virtual doublon enters: it changes only the antisymmetric pseudospin channel. With $\Pi_t = (I + P_{12})/2$ and $\Pi_s = (I - P_{12})/2$,

$$\begin{aligned} \mathcal{A}_{21} &= R_{12}(p,q)\mathcal{A}_{12}, \\ R_{12}(p,q) &= \Pi_t + \frac{\Delta\mathcal{B} - i\,2U\Gamma_A}{\Delta\mathcal{B} + i\,2U\Gamma_A}\Pi_s \\ &= \frac{\Delta\mathcal{B}\, I_{12} + i\,2U\Gamma_A P_{12}}{\Delta\mathcal{B} + i\,2U\Gamma_A}. \end{aligned} \tag{H19}$$

Thus the contact matrix follows directly from the two projector-resolved matching equations; no rank-one Feshbach ansatz is added.

For $U \neq 0$, set

$$u(p) = \frac{\mathcal{B}(p)}{2U\Gamma_A}, \qquad x = u(p) - u(q). \tag{H20}$$

Then

$$R_{12}(x) = \frac{xI_{12} + iP_{12}}{x + i} = \frac{1}{x+i} \begin{pmatrix} x+i & 0 & 0 & 0 \\ 0 & x & i & 0 \\ 0 & i & x & 0 \\ 0 & 0 & 0 & x+i \end{pmatrix}. \tag{H21}$$

The bare pair-energy factor is absorbed exactly because

$$\mathcal{B}(p) - \mathcal{B}(q) = (p - q)(p + q - 2\epsilon_d - U). \tag{H22}$$

The rapidity map is generically two-to-one, $u(p) = u(2\epsilon_d + U - p)$, and is singular at $U = 0$. Global injectivity is not required for the matrix algebra, while the $U = 0$ quadratic benchmark is treated separately.

For the unnormalized matrix $\mathcal{R}_{12}(x) = xI_{12} + iP_{12}$, the permutation identities give

$$\begin{aligned} & \mathcal{R}_{12}(u - v)\mathcal{R}_{13}(u - w)\mathcal{R}_{23}(v - w) \\ & \quad = \mathcal{R}_{23}(v - w)\mathcal{R}_{13}(u - w)\mathcal{R}_{12}(u - v), \end{aligned} \tag{H23}$$

and, for every $S \in GL(2, \mathbb{C})$,

$$(S \otimes S)^{-1}\mathcal{R}_{12}(x)(S \otimes S) = \mathcal{R}_{12}(x), \tag{H24}$$

because $[P_{12}, S \otimes S] = 0$. For $S = S_C = e^{\theta\sigma_y/2}$, $\tanh\theta = G/D$,

$$S_C^{-1}(D\sigma_z + iG\sigma_x)S_C = \sqrt{D^2 - G^2}\,\sigma_z. \tag{H25}$$

The same canonical transformation on bath and impurity fields leaves the kinetic term, scalar hybridization, and $n_{d+}n_{d-} = N_d(N_d - 1)/2$ invariant. It therefore preserves the ordinary YBE rather than producing a new bulk R-matrix.

The non-diagonal pseudospin sector can be displayed without choosing the diagonal positive-metric frame. Define

$$\mathcal{J}^\mu = \frac{1}{2}\int_0^{\mathcal{L}} dx\, \bar{\chi}\sigma_\mu\chi \frac{1}{2}\bar{d}\sigma_\mu d, \qquad \mathcal{J}^\pm = \mathcal{J}^x \pm i\mathcal{J}^y. \tag{H26}$$

For $M = D\sigma_z + iG\sigma_x$, the conserved global charge is

$$Q_M = 2D\mathcal{J}^z + iG(\mathcal{J}^+ + \mathcal{J}^-). \tag{H27}$$

Accordingly, the nonzero ladder terms (often written $S^{\pm}$) are not discarded. For $D^2 \neq G^2$, the global canonical action induced by S_C maps Eq. (H27) to $2\sqrt{D^2 - G^2}\,\mathcal{J}^z$, while Eq. (H24) keeps the same Yang–Baxter algebra.

At $D^2 = G^2$, the diagonalizer in Eq. (H25) is singular, but $M^2 = 0$. On a finite ring the bounded gauge map moves the constant matrix into

$$\mathcal{G} = e^{iM\mathcal{L}/v} = I + iM\mathcal{L}/v, \qquad \mathcal{G}^{-1} = I - iM\mathcal{L}/v. \tag{H28}$$

Because Eq. (H24) also holds for this non-diagonalizable $\mathcal{G}$, the monodromy

$$T_a(z) = \mathcal{G}_a \mathcal{R}_{aN}(z - u_N) \cdots \mathcal{R}_{a1}(z - u_1) \tag{H29}$$

satisfies twisted RTT and generates commuting transfer matrices. The finite-volume domain proof, many-body Jordan theorem, projective descendant-spin-root contraction, and four-state Fock lift are proved in the companion work [10]. There the EP value of Eq. (H27) is treated directly as a nonzero nilpotent global ladder generator: it creates the symmetry-protected Jordan chain rather than acting as an omitted integrability-breaking perturbation. These results are quoted here only to identify the precise continuation of the physical contact algebra.

The supplied script `verify_finiteU_contact_algebra.py` independently checks the displayed 4×4 matrix. For generic complex rapidities and a generic non-unitary S, it gives Frobenius residuals 1.14×10^{-15} for YBE, 4.48×10^{-16} for $GL(2,\mathbb{C})$ covariance, and 1.66×10^{-16} for the metric diagonalization; the singlet and triplet eigenvalue residuals vanish to machine precision.

5. One-particle contact-phase quantization

Inside the passivity window, $S^\dagger S = I$. Let $\mathsf{s}_a = e^{2i\delta_a}$, $a = 1, 2$, denote its two eigenvalues. These scattering eigenchannels are not identified with the fixed chiral labels. Periodic closure on a ring after the stated finite-window radial linearization gives the exact one-particle phase condition

$$\det\big[I - e^{ik\mathcal{L}} S(k)\big] = 0 \quad \Longleftrightarrow \quad e^{ik_{ja}\mathcal{L}} \mathsf{s}_a(k_{ja}) = 1. \tag{H30}$$

This is exact phase counting for the complete causal one-particle matrix. It is logically distinct from the multiparticle contact matrix Eq. (H19), which requires the additional controlled linearization stated above.

Appendix I: Biorthogonal Structure and Eigenvectors

The frozen impurity block is $h_d = \left[\begin{smallmatrix} \tilde{\epsilon}_\xi + D & iG \\ iG & \tilde{\epsilon}_\xi - D \end{smallmatrix}\right]$ with splitting $s = \sqrt{D^2 - G^2}$ and eigenvalues $E_\pm = \tilde{\epsilon}_\xi \pm s$.

a. Right eigenvectors.

$$|r_+\rangle \propto \begin{pmatrix} G \\ i(D - s) \end{pmatrix}, \quad |r_-\rangle \propto \begin{pmatrix} G \\ i(D + s) \end{pmatrix}. \tag{I1}$$

b. Left eigenvectors. Because h_d is complex symmetric, the left eigenvectors may be taken as the transposes of the right eigenvectors:

$$\langle l_+| \propto \begin{pmatrix} G & i(D - s) \end{pmatrix}, \quad \langle l_-| \propto \begin{pmatrix} G & i(D + s) \end{pmatrix}. \tag{I2}$$

Direct verification gives $\langle l_\pm | r_\mp \rangle = 0$ and

$$\langle l_+|r_+\rangle = 2s(D - s), \qquad \langle l_-|r_-\rangle = -2s(D + s). \tag{I3}$$

Thus the biorthogonal normalization diverges as $1/s$ at the EP:

$$\mathcal{M}_\pm \mathcal{N}_\pm = \frac{1}{\langle l_\pm | r_\pm \rangle} \sim \frac{1}{s} \xrightarrow{s \to 0} \infty. \tag{I4}$$

This is the normalization of an individual eigenspace, not an additional physical spectral weight. The rescaling cancels within each normalized projector, and the divergent pieces of the two projectors cancel in the complete resolvent [Eq. (G9)]. It is consistent with $\kappa_{\mathrm{imp}} \sim C/|s|$ [Eq. (A2)]. At the EP ($s \to 0$): $|r_+\rangle \propto |r_-\rangle$, eigenvectors coalesce, and the Jordan block emerges with $h_d|r_+\rangle = E_{\mathrm{EP}}|r_+\rangle$, $(h_d - E_{\mathrm{EP}})|j\rangle = |r_+\rangle$.

1. Explicit Contact Projector

The normalized rank-one projector in the $\{\tilde{d}_+, \tilde{d}_-\}$ contact space is

$$P_+ = \frac{|r_+\rangle\langle l_+|}{\langle l_+|r_+\rangle} = \frac{1}{2s(D - s)} \begin{pmatrix} G^2 & iG(D - s) \\ iG(D - s) & -(D - s)^2 \end{pmatrix}, \tag{I5}$$

with $P_+^2 = P_+$. The full Hamiltonian $\tilde{H}(k)$ contains additional diagonal bath dispersions $\varepsilon_{c,\pm}(k)$ that do not enter this local projector. The projector is used only to track biorthogonal normalization; by itself it does not establish a two-body R-matrix for the full energy-dependent channel-converting problem. The distinct finite-U contact derivation under controlled linearization is given in Sec. H 4.

Appendix J: Causal Fisher–Lee and Wigner–Smith Check

For the compensated Markovian small-width kernel in the spin representation, define

$$H_s = D\sigma_x - i\bar{\Gamma} I + iG\sigma_z, \qquad G_s^R = (\omega I - H_s)^{-1}, \tag{J1}$$

$$\Gamma_s = 2(\bar{\Gamma} I - G\sigma_z). \tag{J2}$$

The passivity condition $\bar{\Gamma} \geq |G|$ makes Γ_s positive semidefinite. The scattering and delay matrices are

$$S_s = I - i\Gamma_s^{1/2} G_s^R \Gamma_s^{1/2}, \tag{J3}$$

$$Q_s = -iS_s^\dagger \partial_\omega S_s. \tag{J4}$$

The chiral matrices follow by the simultaneous rotation in Eq. (H7). Direct use of the Dyson identity gives $S_\nu^\dagger S_\nu = I$ and $Q_\nu = Q_\nu^\dagger$ in either representation $\nu = s, \chi$. The trace also obeys

$$\operatorname{Tr} Q_\nu = -i\,\partial_\omega \log\det S_\nu. \tag{J5}$$

In the compensated unbroken Markov core, let $x = \omega - \tilde{\epsilon}_\xi$ and $s = \sqrt{D^2 - G^2} \in \mathbb{R}$. Causality fixes the scattering determinant entirely from the two poles and their advanced zeros:

$$\det S(\omega) = \prod_{\alpha=\pm} \frac{x - \alpha s - i\bar{\Gamma}}{x - \alpha s + i\bar{\Gamma}}, \tag{J6}$$

$$\operatorname{Tr} Q(\omega) = \sum_{\alpha=\pm} \frac{2\bar{\Gamma}}{(x - \alpha s)^2 + \bar{\Gamma}^2}. \tag{J7}$$

At the resonance center this gives the analytic pre-EP concentration

$$\operatorname{Tr} Q(\tilde{\epsilon}_\xi) = \frac{4\bar{\Gamma}}{s^2 + \bar{\Gamma}^2}. \tag{J8}$$

Relative to the matched Hermitian kernel,

$$\frac{\operatorname{Tr} Q_G(\tilde{\epsilon}_\xi)}{\operatorname{Tr} Q_{G=0}(\tilde{\epsilon}_\xi)} = \frac{D^2 + \bar{\Gamma}^2}{D^2 - G^2 + \bar{\Gamma}^2} > 1 \tag{J9}$$

for $0 < |G| < |D|$ at fixed D and $\bar{\Gamma}$. This is a basis-independent analytic concentration of the scattering phase density along the specified matched path: the two fixed-weight phase peaks move together and concentrate at the center. Its integral remains two, and changing D or the rate eigenvalues provides adverse controls, so Eq. (J9) is not a universal enhancement theorem or a Kondo-temperature formula. Resolving the determinant into the two eigenvalues s_a of the complete unitary matrix gives

$$\frac{1}{2\pi} \operatorname{Tr} Q_\nu = \sum_{a=1}^{2} \frac{1}{2\pi i} \partial_\omega \log \mathsf{s}_a \equiv K_{\mathrm{ph}}. \tag{J10}$$

Thus the logarithmic contact-phase density of Eq. (84) is not an independent scale ansatz: it is the causal scattering density-of-states shift of the same complete retarded/advanced problem. At $\beta_0 = 0.50$, the independent full-matrix gate gives a maximum scattering-eigenvalue modulus error 9.99×10^{-16} and a finite-difference phase–Smith identity error 5.53×10^{-9}. The basis-resolved coefficients are given in Eq. (H9); no individual off-diagonal matrix element is used as a basis-independent diagnostic.

At the EP, $D = G$ and the pole pair coalesces at $E_{\mathrm{EP}} = -i\bar{\Gamma}$. The complete Jordan resolvent gives

$$\operatorname{Tr} Q_{\mathrm{EP}}(\omega) = \frac{4\bar{\Gamma}}{\omega^2 + \bar{\Gamma}^2}, \qquad \int \frac{d\omega}{2\pi} \operatorname{Tr} Q_{\mathrm{EP}} = 2. \tag{J11}$$

For $\bar{\Gamma} = 0.12$, this predicts $\operatorname{Tr} Q_{\mathrm{EP}}(0) = 33.3333333$, reproduced numerically to machine precision. The finite-window numerical integral is 1.99236; the small deficit is the omitted Lorentzian tail. Across the validation grid, the maximum scattering-unitarity error is 1.33×10^{-15}, the maximum Smith-Hermiticity error is 2.52×10^{-14}, and the maximum discrepancy in Eq. (J5) is 4.97×10^{-14}.

Figure S4 includes the essential adverse control. A rate-matched off-EP kernel with $D = 0.05$ reaches a largest proper-delay peak 63.77, exceeding the EP value 30.56. Once normalized by the slowest decay rate, the EP is not enhanced. The delay profile is therefore used as a causal coalescence and linewidth fingerprint, not as an independent many-body amplification factor.

Appendix K: Causal Channel Density and Matrix Poor-Man's Scaling

This appendix records the complete Green-function matrix analysis used for Supplemental Fig. S5. It is the standard one-loop poor-man's-scaling construction [11] applied to the energy-dependent positive channel density of the projected kernel.

1. Positive matrix density

For either the spin or chiral representation,

$$\begin{aligned}\rho_\nu(\omega) &= \frac{i}{2\pi}\left[G_\nu^R(\omega) - G_\nu^A(\omega)\right] \\ &= \frac{1}{2\pi} G_\nu^R(\omega)\Gamma_\nu(\omega) G_\nu^A(\omega), \qquad \nu = s, \chi.\end{aligned} \tag{K1}$$

The second equality is the retarded/advanced Dyson identity. It proves positivity directly:

$$v^\dagger \rho_\nu v = \frac{1}{2\pi} \left\| \Gamma_\nu^{1/2} G_\nu^A v \right\|^2 \geq 0. \tag{K2}$$

It also makes basis covariance explicit, $\rho_\chi = U_H^\dagger \rho_s U_H$. Consequently the two density eigenvalues are independent of whether the calculation is displayed in the spin or fixed chiral basis.

2. Exchange matrix and flow

The finite-U Schrieffer–Wolff transformation fixes the reciprocal bare exchange matrix,

$$J_0 = 2W^2 \left(\frac{1}{|\epsilon_d|} + \frac{1}{|\epsilon_d + U|} \right) I. \tag{K3}$$

Let $\ell = \ln(D_{\rm uv}/\Lambda)$ and define the symmetric shell density

$$\rho_{\rm sh}(\Lambda) = \rho(+\Lambda) + \rho(-\Lambda). \tag{K4}$$

The noncommuting one-loop flow integrated in the numerical gate is

$$\frac{dJ}{d\ell} = J(\ell)\rho_{\rm sh}(\Lambda)J(\ell). \tag{K5}$$

No simultaneous diagonalization of J and $\rho_{\rm sh}$ is assumed. For Hermitian J and positive Hermitian $\rho_{\rm sh}$, $J\rho_{\rm sh}J$ is Hermitian, so the flow preserves Hermiticity. Under a fixed unitary basis change, $J \mapsto U^\dagger JU$ and $\rho \mapsto U^\dagger \rho U$, making Eq. (K5) covariant.

To compare channels without assigning significance to a basis component, the code forms

$$\rho_{\rm av}(\Lambda) = \frac{\rho_{\rm sh}(\Lambda)}{2},$$
$$g(\Lambda) = \rho_{\rm av}^{1/2} J(\Lambda) \rho_{\rm av}^{1/2}, \qquad g_{\max} = \lambda_{\max}[g(\Lambda)]. \tag{K6}$$

The operational scale $T_*(g_*)$ is the cutoff at which $g_{\max}(\Lambda)$ first reaches a specified threshold g_*. It is deliberately called an operational weak-flow scale, not a thermodynamic Kondo temperature.

3. Coefficient and convergence audit

For $k_{\max} = \pi/4$ the gate uses $\rho_{\rm ref} = 1/(2k_{\max})$, $\Gamma_0 = \pi\rho_{\rm ref}W^2 = 0.12$, and therefore

$$W^2 = 0.06, \qquad J_0 = 0.24. \tag{K7}$$

At $\beta_0 = 0.50$, $D_0 = G_0 = 0.10$ and the two rate eigenvalues are 0.04 and 0.44. The selected-grid minimum density eigenvalue is 0.0365034, while the strict pre-EP-window minimum is $0.0241015 > 0$. The maximum difference between the two forms of Eq. (K1) is 1.79×10^{-15}, the exchange Hermiticity error is zero, and the maximum step-refinement drift is 5.92×10^{-10}.

The active/control threshold-scale ratios at $\beta_0 = 0.50$ are 1.0567, 1.0179, 0.9893, and 0.9166 for $g_* = 0.30$, 0.40, 0.50, and 1.00, respectively. The ordering therefore reverses near $g_* = 0.4627$. The calculation establishes a bounded early-flow ordering, but not a threshold-independent enhancement of the strong-coupling Kondo scale.

Appendix L: Finite-U Biorthogonal NRG Residue Audit

We next test the Jordan response by an operator-resolved non-Hermitian NRG calculation in the scalar-hybridization Wilson-chain control of the projected model [12]. This is a weak-coupling structural control of the local finite-U residue in the early Wilson shells, not a low-energy fixed point or a phase diagram for the full SOC-resolved bath. It therefore neither locates nor excludes the interacting continuation of the full-k causal pole EP. That distinct problem is addressed by the exact finite-chain matrix block-Wilson calculation of Sec. C. The present section instead asks whether the scalar Markov-core transition residue retains the nilpotent matrix direction. The two transition operators are propagated independently as $L^\dagger cR$ and $L^\dagger c^\dagger R$; a no-truncation regression checks their noninteracting matrix elements before the interacting scan.

At the exact Jordan point a complete biorthogonal eigenbasis is singular. We therefore calculate at finite coherent detuning $\delta_{\rm coh} > 0$, track the same two impurity-supported poles through U and $\delta_{\rm coh}$, and only then extrapolate to $\delta_{\rm coh} \to 0$. If

$$G(z) = \frac{W_+}{z - z_+} + \frac{W_-}{z - z_-} + G_{\rm reg}(z), \tag{L1}$$

the coefficient that approaches the Jordan double-pole residue is

$$B = \frac{1}{2}(W_+ - W_-)(z_+ - z_-) = B_0 I + B_x\sigma_x + B_y\sigma_y + B_z\sigma_z. \tag{L2}$$

In the Hadamard/chiral basis used by the Wilson-chain adapter, the compensated EP generator is

$$N_{\rm EP} = \Delta_{\rm eff}\sigma_z + i\Gamma_{\rm PT}\sigma_x, \qquad N_{\rm EP}^2 = 0, \tag{L3}$$

we monitor the Frobenius-angle alignment, normalized trace, and nilpotent component mismatch,

$$A_J = \frac{|\operatorname{Tr}(N_{\rm EP}^\dagger B)|}{\|N_{\rm EP}\|_F \|B\|_F},$$
$$f_{\rm tr} = \frac{|\operatorname{Tr} B|}{\sqrt{2}\|B\|_F}, \qquad \epsilon_N = \frac{|B_x - iB_z|}{|B_x| + |B_z|}. \tag{L4}$$

Supplemental Fig. S7 reports the validated part of the audit. The reference calculation uses $\Lambda = 3$, $z = 0.5$, $N_{\rm keep} = 320$, bath exponent $r = 1$, $\beta_0 = 0.50$, $\Delta_{\rm eff} = \Gamma_{\rm PT} = 0.10$, $0 \le U \le 0.10$, and the early/high-energy NRG iterations $n = 4$–6 in charge sectors $q = \pm 1$. Five detunings between 10^{-5} and 10^{-4} are used. The reliable tracked-pair fraction is 0.993939; over the direct finite-detuning ensemble the median alignment is 0.99999963, the median trace fraction is 8.3567×10^{-4}, and the median nilpotent mismatch is 7.3179×10^{-7}. Thus the finite-U calculation retains the nearly traceless nilpotent matrix direction.

The stronger pole-splitting claims do not pass the same convergence gate. Across iteration, sector, fitting window, and detuning model, the complex interaction-exponent ensemble has median $p = 1.359$ with a 68% spread of 1.991. The proposed complex quartic coefficient also fails its plateau test (median absolute log-slope 0.880 and relative complex scatter 1.118). These failures do not prove the absence of a scaling regime, but they preclude extracting a universal exponent, quartic coefficient, or interaction-induced EP floor from this calculation. Accordingly only the matrix-residue result is used in the main text.

Appendix M: Scope and Division of Labor

Three logically distinct statements are now kept separate.

(i) For the complete frozen retarded kernel derived from the driven model, the exact claims are the matrix Fisher–Lee relation, one-particle contact-phase quantization, and the Wigner–Smith identity. These require no many-body factorization.

(ii) After equal-velocity branchwise linearization and folding, with constant M, scalar hybridization, and component-symmetric local U, the exact two-electron contact matrix is Eq. (H19). Its dressed-rapidity difference form and $GL(2,\mathbb{C})$ covariance prove YBE and the compatibility of the metric similarity. On this submanifold the standard interaction does not lift the symmetry-protected EP.

(iii) Ref. [10] supplies the mathematical continuation: the bounded finite-volume gauge equivalence, RLL and twisted RTT relations, commuting transfer matrices, the explicit treatment of the nonzero global pseudospin ladders, many-body Jordan blocks, and the compatible four-state Fock lift. Its descendant *spin* roots meet at the projective point $\lambda = \infty$, with scaled coordinates given by a generalized Laguerre polynomial; no finite Anderson charge-root coalescence is asserted. Those results are not rederived from the physical scattering plots in this article.

Curvature, unequal folded velocities, energy-dependent $M(\omega)$, nonscalar hybridization, or component-dependent interactions break one or more hypotheses of item (ii). They can shift or split the EP and may spoil factorization. No Yang–Baxter theorem or thermodynamic Bethe ansatz is claimed for that complete driven kernel. Conversely, no universal interaction-induced avoided-coalescence scale is introduced: such a scale would require a specified symmetry-breaking self-energy calculation.

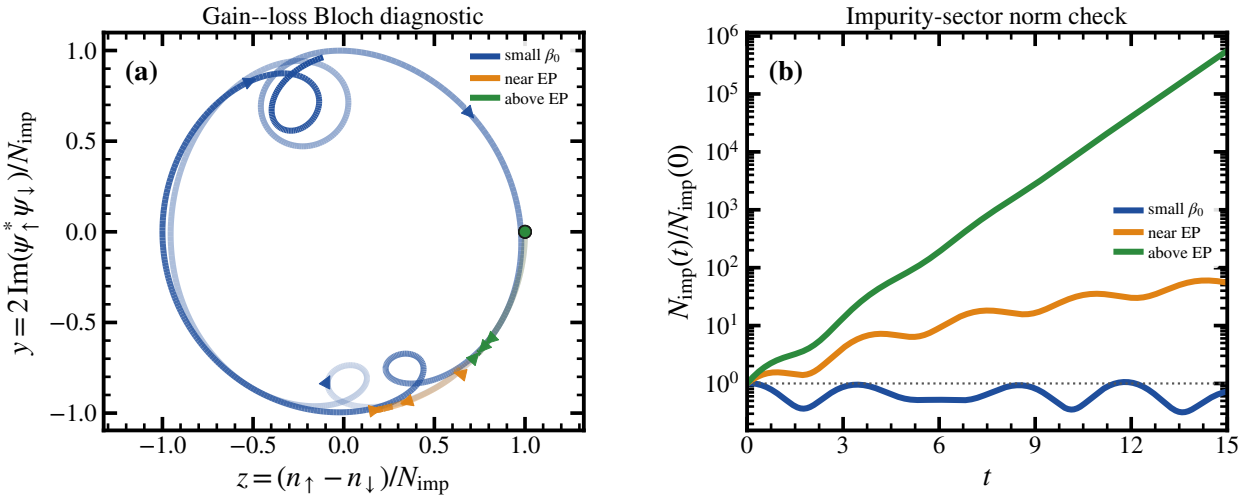


FIG. S1. **Gain–loss Bloch diagnostic for the impurity sector.** This is the same normalized $k = 0$ constant-core calculation as main-text Fig. 2: $\tilde{\epsilon}_\xi = -1$, $D = 0.50$ in panels (a)–(c), $D = 0$ in panels (d)–(f), $G = |\beta_0|$, and fixed $W = 0.75$. It contains no physical temperature, interaction, DOS, or cutoff. Panel (a) shows the normalized (z, y) trajectory for representative values of β_0 below, near, and above the EP. Below the EP the normalized co-decaying trajectory remains close to the unbroken relative-kernel orbit, while close to and beyond the EP the trajectory is progressively distorted. Panel (b) tracks the impurity-sector norm $N_{\mathrm{imp}}(t)/N_{\mathrm{imp}}(0)$ and shows that the above-EP trajectory acquires a strong late-time growth in the normalized diagnostic, whereas the below-EP and near-EP traces remain well controlled over the plotted time window.

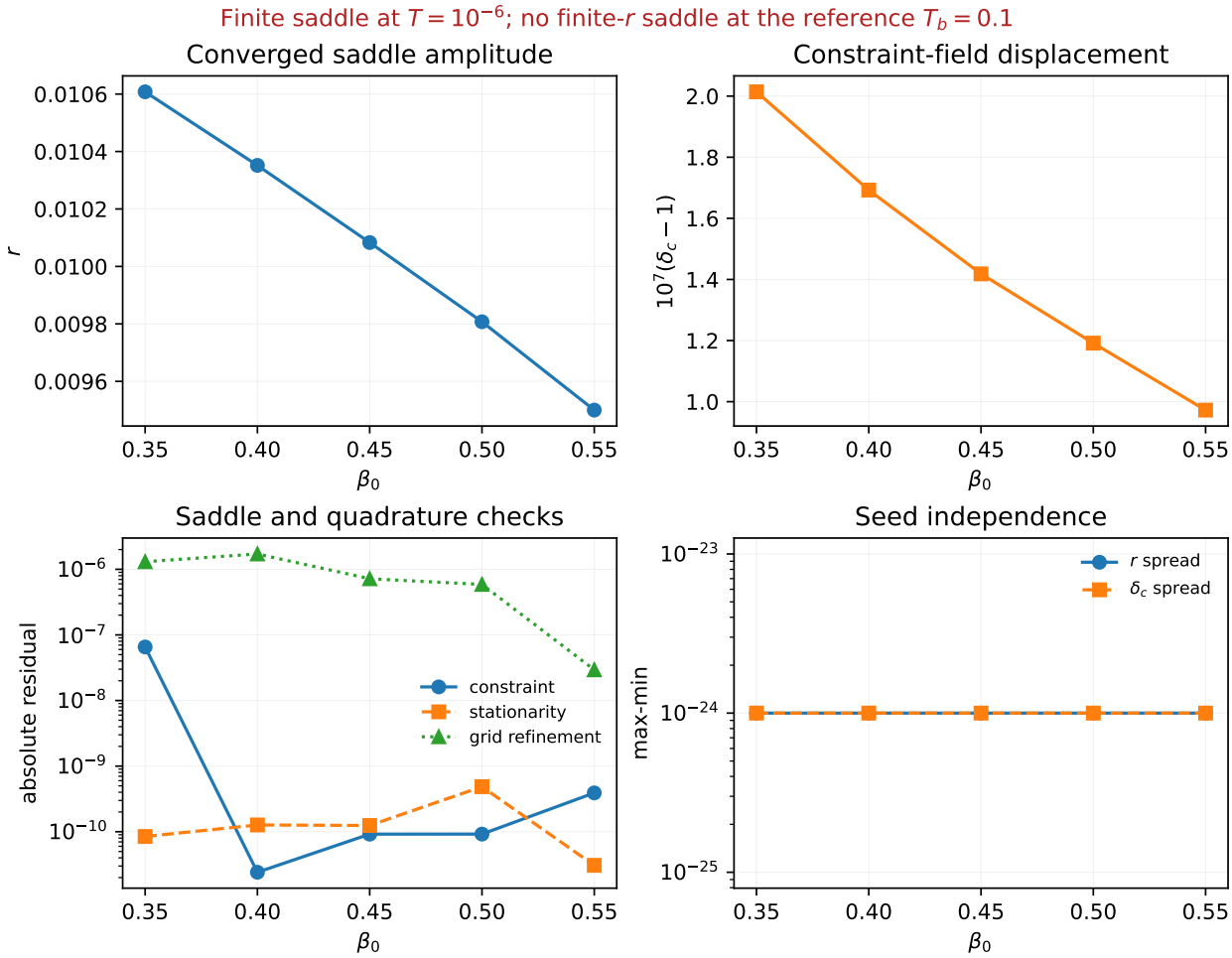


FIG. S2. **Low-temperature $U \to \infty$ saddle and numerical validation tests.** At $T_\mathrm{b} = 10^{-6}$, panel (a) shows the converged amplitude r and panel (b) the constraint-field displacement over $0.35 \le \beta_0 \le 0.55$. Panel (c) reports the final constraint, stationarity, and grid-refinement residuals; panel (d) shows that real, negative-real, and complex initial boson seeds converge to the same gauge-fixed solution. The complete sandwich construction satisfies charge continuity to 3.66×10^{-23}; the maximum constraint, stationarity, and grid-refinement residuals are 6.54×10^{-8}, 4.86×10^{-10}, and 1.71×10^{-6}, respectively. The title records the distinct result at the manuscript reference temperature: the same closure has no finite-r branch at $T_\mathrm{b} = 0.1$. The figure therefore validates the low-temperature saddle and the exact core-EP scaling, but does not claim finite-temperature condensate enhancement.

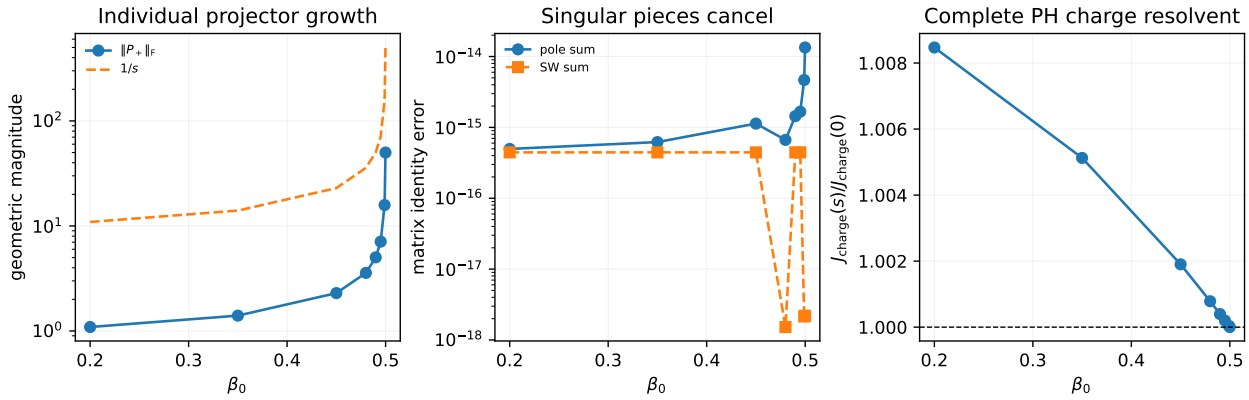


FIG. S3. **Biorthogonal-projector and finite-U resolvent checks.** Panel (a) shows the growth of an individual normalized projector and the artificial $1/s$ factor on the pre-EP side. Panel (b) verifies numerically that the two-projector pole sum equals the direct matrix resolvent and that the two virtual charge sectors equal the closed Schrieffer–Wolff resolvent; the plotted discrepancies are numerical roundoff. Panel (c) shows the complete particle–hole charge factor normalized to its EP limit. It remains finite and approaches unity although each projector grows. These checks exclude an independent Petermann or Z^{bio} multiplier from the Green function and screening estimate.

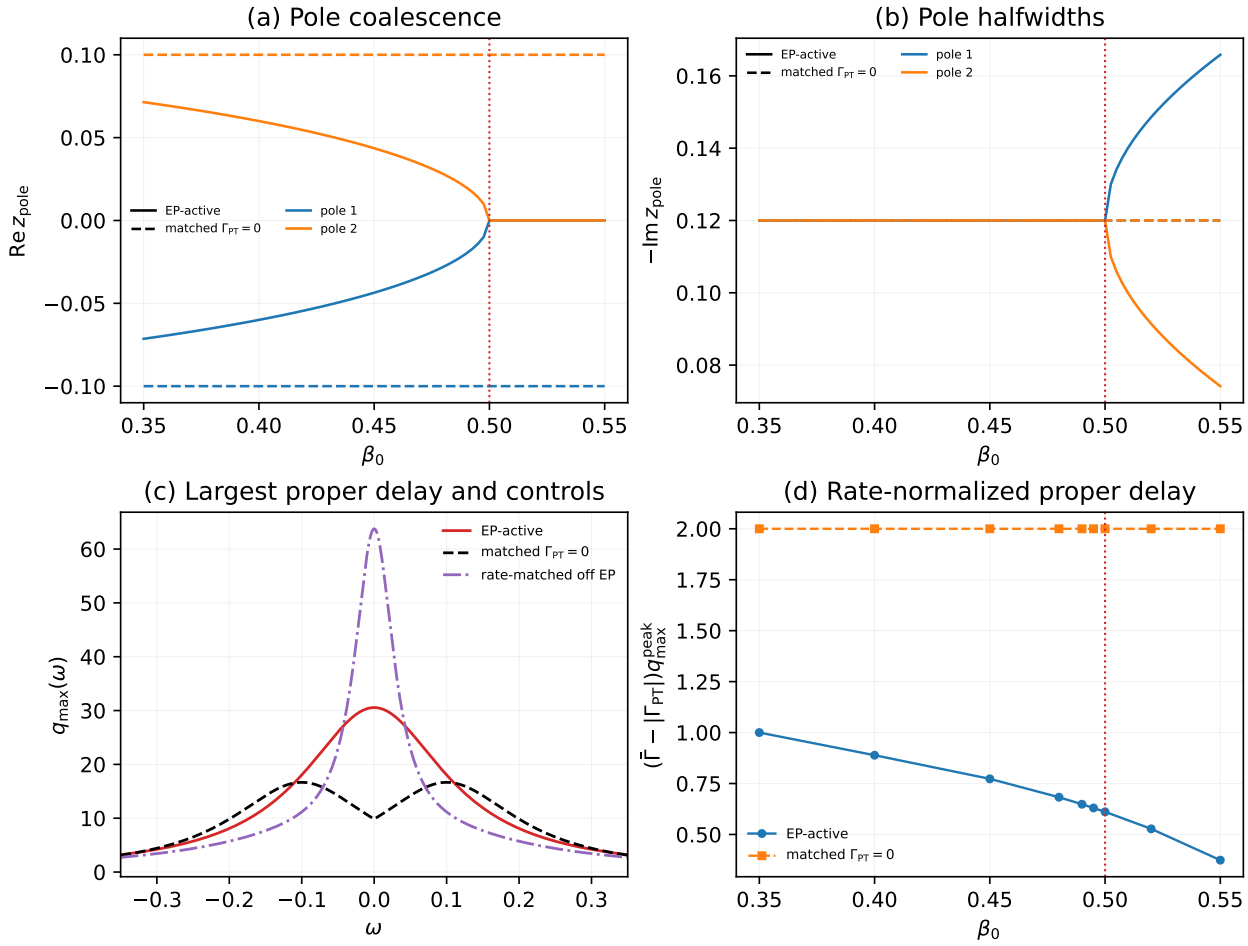


FIG. S4. **Causal Smith-delay validation and adverse controls.** Panel (a) tracks the analytic pole pair to the exact core EP at $\beta_0 = 0.50$. Panel (b) shows the two passive halfwidths: they remain positive throughout the retained window and split only on the relative-broken side. Panel (c) compares the largest proper-delay peak with a Hermitian control matched by total rate and with an off-EP rate-matched kernel ($D = 0.05$); the latter exceeds the EP peak. Panel (d) normalizes the largest proper delay by the slowest rate and likewise shows no independent EP enhancement. Together with the unitarity and trace-delay checks in App. J, these controls establish the Smith profile as a causal coalescence/linewidth fingerprint, not a Kondo-scale multiplier.

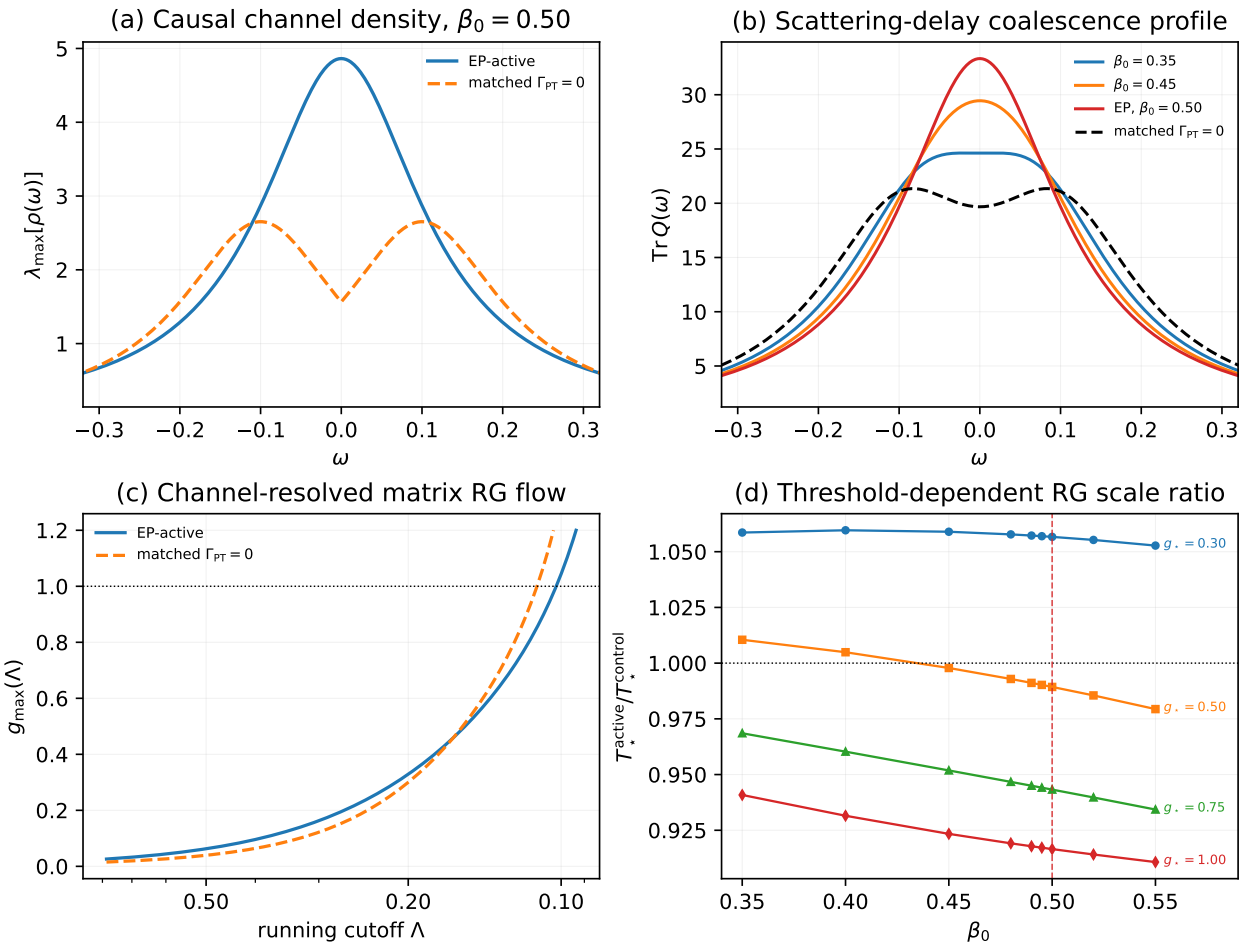


FIG. S5. **Causal scattering and channel-resolved RG support.** The conservative fixed-W validation uses the complete positive channel density from G^R and G^A, with $\Gamma_0 = 0.12$ and unit charge gap. Panel (a) compares the largest causal channel-density eigenvalue at $\beta_0 = 0.50$ with the matched $\Gamma_{\mathrm{PT}} = 0$ control. Panel (b) gives the Wigner–Smith trace; the passive Fisher–Lee matrix is unitary to 1.33×10^{-15} and $\mathrm{Tr}\,Q(0) = 4/\Gamma_0 = 33.3333$ at the EP. Panels (c) and (d) show that the EP-active exchange eigenvalue is initially larger, but its threshold-scale ratio crosses below the control before strong coupling. This supports a bounded early-flow ordering, not an enhanced thermodynamic Kondo temperature.

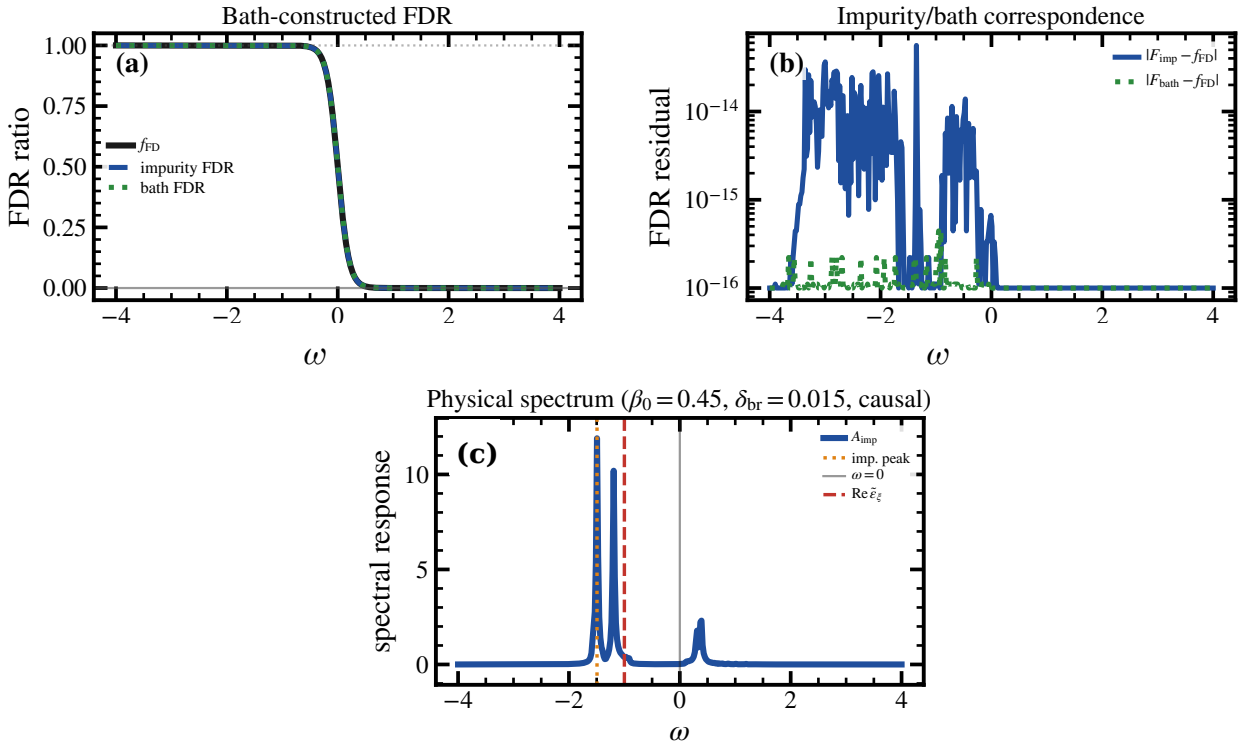


FIG. S6. **Bath-controlled fluctuation–dissipation and spectral checks.** The uniform reference is $\epsilon_\xi = -U/2 = -1$, $U = 2$, and $T_\mathrm{b} = 0.1$. Panels (a) and (b) verify the physical construction $G^<_\mathrm{phys} = G^R\Sigma^<_\mathrm{bath}G^A = f_\mathrm{FD}(G^A - G^R)$ with maximum residual 5.61×10^{-14} at $\delta_{\mathrm{br,FDR}} = 10^{-3}$. Panel (c) shows the physical impurity spectral response with $\delta_\mathrm{br} = 0.015$, evaluated from the complete retarded and advanced kernel. The two equivalent density-of-states evaluations in Eq. (E4) agree numerically.

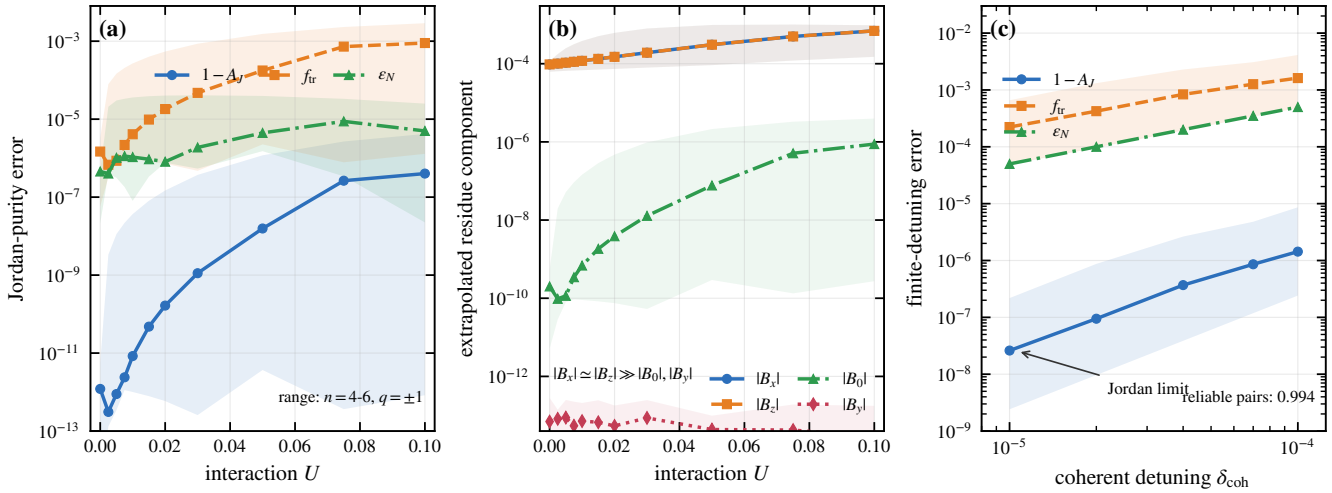


FIG. S7. **Finite-U Jordan transition-residue audit for the scalar-hybridization NRG control.** (a) Zero-detuning extrapolations of the alignment error $1 - A_J$, trace fraction f_tr, and nilpotent mismatch ϵ_N. Curves are medians and bands are the full range over the early/high-energy iterations $n = 4$–6 and charge sectors $q = \pm 1$. (b) Pauli components of the extrapolated double-pole coefficient B: $|B_x| \simeq |B_z| \gg |B_0|, |B_y|$, as required by $N_\mathrm{EP} = \Delta_\mathrm{eff}\sigma_z + i\Gamma_\mathrm{PT}\sigma_x$. (c) Direct finite-detuning convergence; curves are medians and bands are the 16th–84th percentile over U, iteration, and tracked sector. The reliable-pair fraction is 0.994. Parameters are $\Lambda = 3$, $z = 0.5$, $N_\mathrm{keep} = 320$, $r = 1$, $\beta_0 = 0.50$, $\Delta_\mathrm{eff} = \Gamma_\mathrm{PT} = 0.10$, and $0 \le U \le 0.10$. The figure establishes persistence of the nilpotent matrix-residue direction in this weak-coupling structural window; it does not establish a low-energy fixed point, pole coalescence, a universal interaction law, a thermodynamic Kondo scale, or a phase boundary.

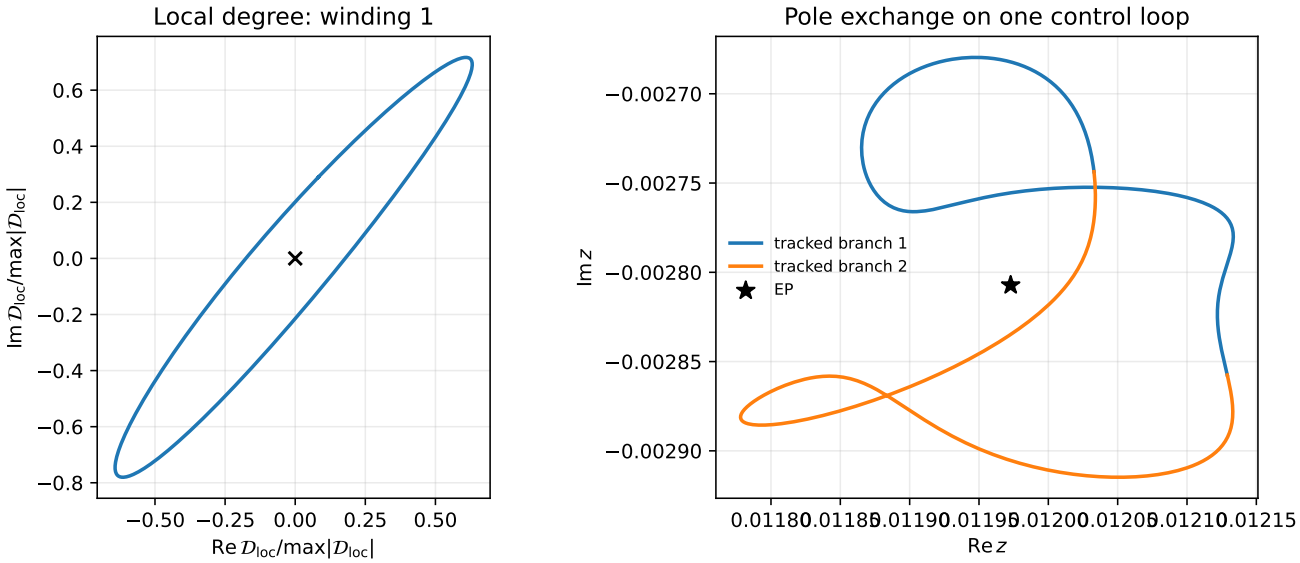


FIG. S8. **Full-momentum two-cut nonlinear causal-pole certificate.** The stationary rotating-drive witness uses $\lambda = 0.05$ and fixed $W = 0.10$, $K = \pi/4$, $\epsilon_O = 120$, $\Omega = 60$, and the solution in Eq. (B34). Left: the normalized local discriminant $\mathcal{D}_{\rm loc} = F(z_c)$ winds once around the origin on a nonsingular ellipse in (ϵ_d, V_O). Right: the two resonance poles exchange after the same control loop; the star marks $z_{\rm EP}$. The winding is $+1$, the pole-separation half-winding is $+1/2$, and all pole solves pass the separation and residual guards. This certifies a nonlinear pole EP of the stated full curved retarded kernel; it is not a certificate for the nilpotent Markov-core EP or for full-momentum Yang–Baxter integrability.

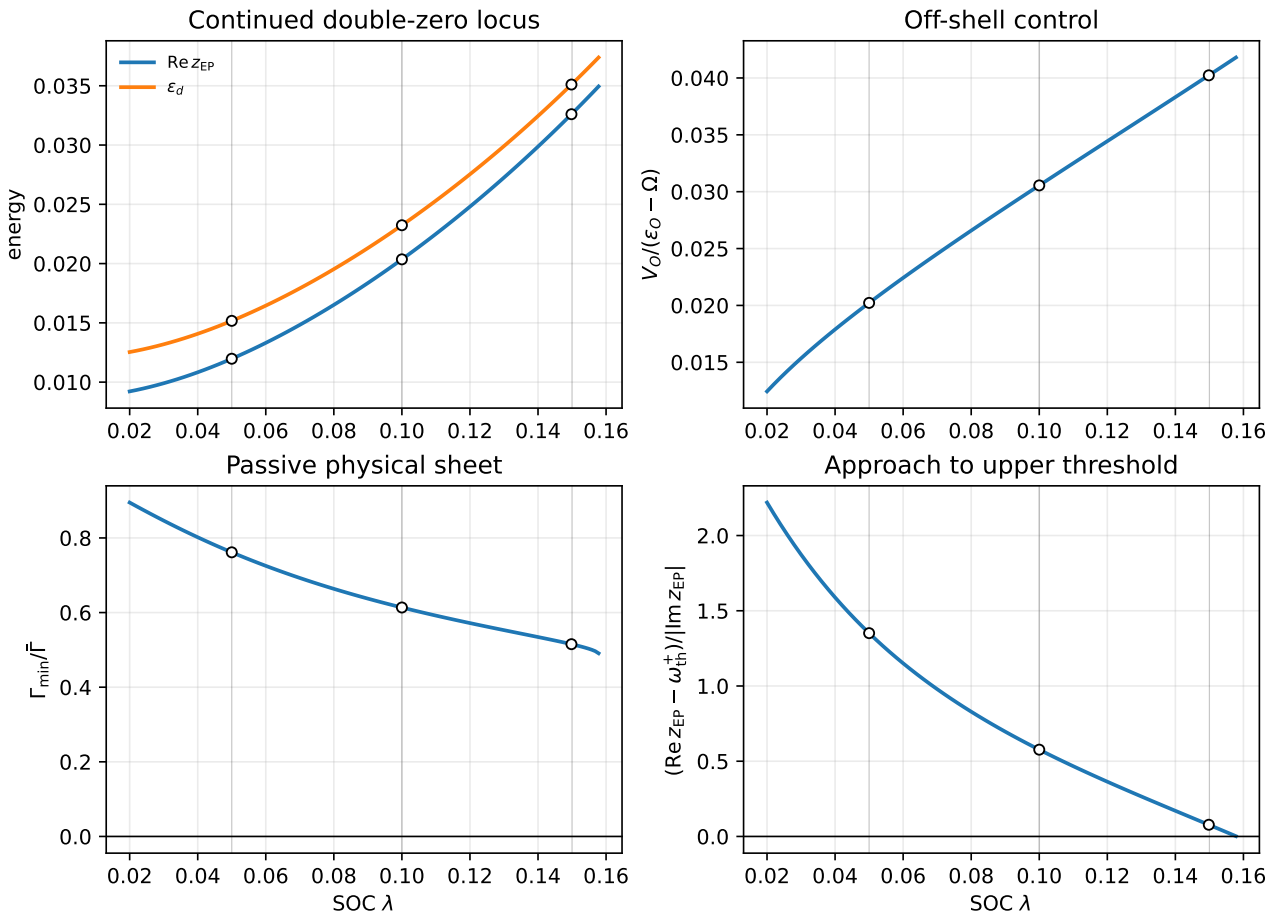


FIG. S9. **Finite SOC family of full-momentum nonlinear pole EPs.** The double-zero equations are continued in λ with $W = 0.10$ fixed, while the real impurity gate ϵ_d and off-shell auxiliary vertex V_O are retuned. Vertical guides mark $\lambda = 0.05$, 0.10, and 0.15, where independent local-winding and pole-exchange certificates give winding $+1$, pole permutation (12), and half-winding $+1/2$. The off-shell ratio remains below 0.042 and the physical rate matrix remains positive. The last panel shows the approach to the driven upper $k = 0$ threshold. This is a one-dimensional locus in the enlarged parameter space; the approach to its threshold endpoint occurs without loss of passivity, and neither that endpoint nor any two-dimensional EP phase region is claimed.

REPRODUCIBILITY

The numerical scripts, parameter tables, and exported data used for the main and Supplemental figures are included in the accompanying archive and in the version-controlled GitHub repository `VMKPHYSMATH/emergent-pt-dirac-impurity` [13]. The current public snapshot is tagged `v2.0.1`; the all-versions record is doi:10.5281/zenodo.21434682. The full-momentum certificate introduced in this revision is included in the submission archive and will be incorporated into the next public archival snapshot. The accompanying source also includes `verify_finiteU_contact_algebra.py`, which reproduces the finite-U R-matrix, metric covariance, and YBE residuals quoted above. The script `full_k_two_cut_ep_certificate.py` reproduces the exact rotating-drive auxiliary closure, analytic full-momentum self-energy, two-cut continuation, nonlinear double-zero solve, local winding, and pole exchange of Fig. S8. The companion script `full_k_ep_soc_family.py` continues the double-zero locus in SOC, repeats the full topological certificate at three representative points, and generates Fig. S9 and its tabulated data. The archived `finiteU_fullk_*` JSON summaries record the exact finite-chain interaction path, local winding, pole exchange, equation-of-motion identity, $N_z = 4 \to 8$ center check, and kept-space calibration of Sec. C. The archived `fullk_n_scaling_gate_summary.json` and `FULLK_N_SCALING_AUDIT.md` record the independently optimized $N = 10$ and 12 common-contour gate and Eq. (C4). The script `make_finiteU_fullk_exact_figure.py` reads those summaries and reproduces Fig. 1. The NRG archive additionally contains the independent transition-operator propagation, no-truncation regression, branch-tracking and convergence scripts, the gate summary, and the exported data used in Fig. S7.

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